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AgriSureGIS: A WebGIS-Based Typhoon Parametric Insurance Assessment for PCIC Region X

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TABLE OF CONTENTS

CHAPTER I: INTRODUCTION	1
1.1 Background of the Study	1
1.2. Statement of the Problem	3
1.2.1 General Problem	3
1.2.2 Specific Problem	3
1.3. Objectives of the Study	4
1.3.1 General Objective	4
1.3.2 Specific Objectives	4
1.4. Scope and Delimitation	5
1.4.1 Scope	5
1.4.2 Delimitations	6
1.5. Significance of the Study	7
CHAPTER II: REVIEW OF RELATED LITERATURE	9
2.1. Preliminary Literature Review	9
2.1.1. WebGIS and Spatial Decision Support in Agriculture	9
2.1.2. Eliminating Data Silos and Building Centralized Systems	10
2.1.3. Multi-Criteria Filtering for Spatial and Attribute Data	11
2.1.4. Integration of Meteorological Data for Automated Risk Assessment	12
2.1.5. Automated Meteorological Data Acquisition and Notification	12
2.1.6. Secure Localized Spatial Databases for Information Management	13
2.1.7. Hybrid Deployment Progressive Web Applications (PWA) and Native Frameworks	13
2.1.8. A WebGIS Ecosystems Utilizing Google Earth Engine	14
2.1.9. Synthesis of the Reviewed Literature	15
2.2.10 Comparison of Existing Systems/Studies	15
Gantt Chart	17
CHAPTER III: METHODOLOGY	18
3.1. Software Development Model	18
3.2. Planning and Requirement Analysis	19
3.2.1 Information Gathering	19
3.2.2 Software Requirements Specifications	22
3.3. System Design	27
3.3.1 System Architecture	27
3.3.2 Use Case Diagram	28
3.3.3 Context Diagram	29
3.3.4 Data Flow Diagram	31
3.3.5 Entity Relationship Diagram	36
3.3.5.1 Database Dictionary	37
3.3.6 Parametric Indemnity Process Flowchart	47

3.4. System Development	54
3.4.1 Hardware Requirements	54
3.4.2 Software Requirements	55
3.5. System Testing and Evaluation	61
3.5.1 Functionality Testing	61
3.5.2 Parametric Calculation and Spatial Accuracy Testing	61
3.5.3 System Usability Scale Testing	63
3.6. Wireframe	64
3.6.1 Low Fidelity Wireframe	64
3.7. Deployment	69
References	70

LIST OF TABLES

Table 1. Comparative Analysis of Related Works	15
Table 2. Database Dictionary	37
Table 3. Hardware Requirements	54
Table 4. SUS Questionnaires	63

LIST OF FIGURES

Figure 1. Gantt Chart	17
Figure 2. Agile Development Model	18
Figure 3. System Architecture	27
Figure 4. Use Case Diagram	28
Figure 5. Context Diagram	29
Figure 6. Data Flow Diagram	31
Figure 7. Entity Relationship Diagram	36
Figure 8. Parametric Indemnity Process Flowchart	47

CHAPTER I

INTRODUCTION

1.1 Background of the Study

Geographic Information Systems (GIS) are widely used around the world to manage and analyze location-based data. In agriculture, GIS helps monitor crops, measure land area, assess environmental risks, and support decision-making. A recent systematic review by Raihan (2024) explains that GIS improves agricultural planning, sustainability, and data-driven decision-making by combining spatial and non-spatial information. GIS allows agricultural agencies to transition from basic record-keeping to advanced spatial analysis, enabling them to visualize farmland distributions and assess environmental vulnerabilities with high precision.

In regions highly susceptible to extreme weather events, such as the Philippines, agricultural insurance plays a critical role in managing climate risks and ensuring the economic survival of small-scale farmers. However, traditional damage assessment relies heavily on slow, manual field verification. To overcome this, modern agricultural insurance is shifting toward data-driven, index-based parametric models that utilize real-time meteorological data such as typhoon wind velocities and rainfall indices to automate hazard detection and accelerate claim processing (Nguyen-Huy et al., 2024).

Despite these technological advancements, the Philippine Crop Insurance Corporation (PCIC) Region X faces significant operational bottlenecks due to the persistence of fragmented "data silos." Currently, based on the conducted interview, the agency manages over 500,000+ farmer records using disconnected CSV files, standalone GPX coordinates, and isolated desktop GIS software. This lack of a centralized infrastructure increases manual workloads and hinders rapid, multi-user damage assessments during typhoon emergencies (Abu-AlSondos, 2025). Literature suggests that unified Web-GIS frameworks significantly improve data integration, accessibility, and transparency for insurance systems (Bhakal et al., 2024; Zhang et al., 2024; Rabuh et al., 2024).

The need for modernization is urgent, as PCIC's records are projected to grow significantly over the next five years. While software like QGIS provides powerful analysis, it lacks a customized, web-based interface tailored to PCIC's specific workflow. Furthermore, moving toward unified systems can eliminate the organizational "silos" that slow down risk

assessment and service delivery (Alshareef et al., 2024). Integrating spatial and satellite-based data is now essential for supporting accurate, high-speed insurance programs (Nguyen et al., 2025).

Data silos occur when different parts of an organization store and manage data separately, making it difficult to share information across departments. This can slow decision-making, reduce operational efficiency, and increase costs in organizations, including insurance companies. Alshareef et al., (2024) explain that in the insurance sector, organizational culture and leadership play a significant role in eliminating data silos and adopting business intelligence systems that support integrated data management. Their study in Jordan's insurance sector highlights that modern insurance organizations benefit when data systems are unified, which leads to faster access to information, better risk assessment, and improved service delivery.

Given the global advancements in Web-GIS and geospatial insurance systems and the need to eliminate data silos in insurance data management, there is a clear opportunity to modernize PCIC's data system. Therefore, this study proposes a localized Web-GIS platform for PCIC Region X. The system will centralize farmer profiles and insurance records in a secure, localized spatial database to ensure data privacy (Ilba, 2021). Crucially, the platform will feature an automated disaster response module that extracts real-time weather data from Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) Tropical Cyclone Bulletins. By overlaying typhoon wind signals against the PCIC "Risk Exposure List," the system will rapidly calculate affected farm polygons and potential claims (Chen et al., 2025). Furthermore, the system will support spatial-to-attribute data linking, advanced filtering tools, mapping visualization, and automated PDF report generation. By developing and pilot-testing this localized, data-driven system, the study aims to improve operational efficiency, secure data privacy, and accelerate disaster response in crop insurance management in the Philippines.

1.2. Statement of the Problem

1.2.1 General Problem

The PCIC Region X manages a vast inventory of geospatial data, including farm boundaries (polygons) and agricultural insurance records containing personal information of farmers. However, this data is currently "siloe" within localized files and standalone QGIS software. In the landscape of modern agricultural insurance, this fragmented data architecture creates severe operational bottlenecks. When extreme weather events occur, GIS specialists must manually extract data from PAGASA Tropical Cyclone Bulletins and individually cross-reference typhoon wind signals against localized farm polygons to estimate crop damages. This manual processing is highly inefficient, prone to human error, and causes significant delays in the parametric insurance indemnification process. Furthermore, the agency lacks a centralized spatial platform capable of rapidly filtering risk exposures and generating immediate damage assessment reports.

1.2.2 Specific Problem

Specifically, the study seeks to address the following problems:

- The lack of a centralized spatial database to manage the growing volume of farmer metadata alongside their respective farm lot polygons.
- The absence of an interactive web-based dashboard that allows personnel to rapidly execute spatial filtering of farmer records by municipality, province, or RSBSA ID.
- The inefficiency of manually downloading and extracting real-time weather data and wind signal thresholds from public PAGASA Tropical Cyclone Bulletins.
- The inability to automatically cross-reference typhoon wind signals against farm locations to instantly estimate potential parametric insurance claims based on crop stage and risk exposure.
- The lack of an automated reporting tool to instantly generate and export spatial damage assessment summaries for management review and payout approval.

1.3. Objectives of the Study

1.3.1 General Objective

The general objective of this study is to develop and implement a localized WebGIS platform for the Philippine Crop Insurance Corporation (PCIC) Region X that centralizes spatial farm boundaries and farmer insurance metadata. This system aims to eliminate fragmented data silos, automate the extraction of weather data from PAGASA, and streamline parametric disaster risk assessments to expedite insurance claim settlements.

1.3.2 Specific Objectives

Specifically, this study aims to:

- a) To **design and develop** a centralized spatial database (using PostgreSQL/PostGIS) to efficiently store, manage, and link the growing volume of farmer metadata with their respective farm lot polygons.
- b) To **develop an interactive web-based mapping dashboard** equipped with spatial filtering tools, allowing personnel to instantly isolate and view farmer records by municipality, province, or RSBSA ID.
- c) To **create an automated data extraction module** capable of parsing real-time weather data and typhoon wind signal thresholds directly from public PAGASA Tropical Cyclone Bulletins.
- d) To **build a spatial processing engine** that automatically overlays extracted typhoon wind signals against farm locations to instantly calculate risk exposure and estimate parametric insurance claims based on crop stage.
- e) To **integrate an automated reporting feature** that allows users to instantly generate and export spatial damage assessment summaries into CSV format for management review and payout approval.
- f) To **evaluate the functionality and usability** of the system by conducting functionality testing, accuracy testing and usability testing with an authorized PCIC personnel, ensuring the tool effectively meets their operational requirements for rapid disaster risk assessment.

1.4. Scope and Delimitation

1.4.1 Scope

The development of this AgriSureGIS platform is specifically designed to modernize the spatial data management and insurance verification workflows of the Philippine Crop Insurance Corporation Regional Office X. The scope of the project encompasses the following core components:

This study focuses on developing a centralized, web-based platform that integrates farmer profiles, insurance metadata, and spatial farm boundaries into a secure, localized spatial database rather than a public cloud repository. At its core, the system features an automated disaster response module with a dedicated geoprocessing tool designed to extract real-time weather data from PAGASA Tropical Cyclone Bulletins, overlay typhoon wind signals against the PCIC's Risk Exposure List. The computational scope of this parametric assessment engine is specifically focused on executing hazard duration calculations to estimate potential claims based strictly on the predefined Typhoon-Induced Strong Winds matrix.

To support these workflows, the platform delivers essential GIS functionalities, including seamless spatial-to-attribute data linking and an interactive mapping dashboard equipped with advanced filtering tools to sort data by farmer name, municipality, province, barangay, or RSBSA ID. Furthermore, the study incorporates satellite-based crop verification by utilizing the Google Earth Engine API to fetch and overlay cloud-penetrating Synthetic Aperture Radar imagery. This specific capability provides the internal GIS specialist with a localized spatial visual asset to manually inspect and confirm whether the insured farm boundaries were actively planted with crops prior to the onslaught of a typhoon, effectively streamlining the validation process and bypassing the need for immediate, post-disaster physical site inspections. Structurally, this technological infrastructure and its primary dependencies rely on open-source repositories like GitHub for custom parsers and external API keys like GEE for base mapping, meaning the platform's mapping and data extraction capabilities are intentionally aligned with the availability, rate limits, and continuous support of these external web services.

1.4.2 Delimitations

To ensure the feasibility and successful deployment of the system within the given timeframe, the study is bound by the following delimitations:

This study is subject to several technical limitations, primarily that AgriSureGIS operates strictly as a standalone WebGIS platform without direct API integration into the PCIC's centralized system, relying instead on manual data imports from the legacy PCIC Automated Business System via standardized CSV and GPX files. Furthermore, platform accessibility is confined to an office-based environment for authorized GIS specialists and management using desktop or laptop computers; the system lacks a mobile application and does not support offline field data collection or offline-to-online synchronization for field adjusters. Geographically, the implementation is strictly limited to PCIC Region X of the Philippines. Moreover, the operational scope of the AgriSureGIS system is confined to the platform developed in this study, and the data managed is exclusively for rice crops, excluding all other commodities.

In terms of data processing and analysis, the automated weather extraction module is specifically tailored to the standardized PDF formats of PAGASA Tropical Cyclone Bulletins rather than live API feeds from global meteorological agencies. Additionally, while the platform streamlines daily insurance workflows and rapid damage assessments, it is dependent on external tools and is not intended to replace complex spatial analysis features found in professional desktop GIS software like QGIS or Google Earth Engine. Additionally, the application of Google Earth Engine within this system is strictly delimited to the backend extraction, rendering, and visual display of raw Sentinel-1 SAR imagery for manual interpretation by the GIS specialist. The system completely excludes the development or integration of automated crop classification models, digital image processing algorithms, or machine learning frameworks for unclassified datasets, as processing unclassified remote sensing data through automated classification constitutes an entirely separate study. Finally, regarding parametric accuracy and basis risk, it is beyond the scope of this IT Capstone Project to evaluate or improve the meteorological or actuarial accuracy of the parametric index itself; the platform operates under the assumption that the predefined Risk Exposure matrices provided by the PCIC and the weather data from PAGASA are inherently accurate.

1.5. Significance of the Study

To the PCIC, this study fundamentally re-engineers the institutional capacity of PCIC by replacing an error-prone, bottleneck-heavy administrative process with a resilient, automated data pipeline. By establishing a direct, local system integration that mitigates external downtime vulnerabilities such as the routine data purging cycles of state weather platforms the institution secures a permanent, historical archive of geospatial metadata. Crucially, this system changes the paradigm of post-disaster response. By combining pre-established, local farm polygon databases with automated storm trajectory mapping, PCIC can execute data-driven, predictive damage validation. This eliminates the logistical and financial burden of sending physical adjustment teams to geographically isolated areas after a disaster, shifting the institution from a reactive operational model to an agile, proactive risk management agency.

To the GIS specialists, this system directly resolves severe operational friction for internal technical personnel. Previously, GIS specialists were burdened with low-value, repetitive manual tasks, including the manual retrieval of static documents, manual CSV transcription, and the tedious human tracing of coordinates within standalone software. The automated architecture introduced in this system drastically compresses the time-to-resolution for disaster risk assessments. By automating data ingestion, transformation, and spatial visualization, the time required to calculate municipal risk exposure (hours exposed to specific typhoon wind signals) is reduced from days to minutes. This drastic reduction in labor hours liberates technical specialists to focus on high-level spatial analytics, data verification, and long-term risk modeling, thereby maximizing institutional efficiency and reducing human error during critical disaster response windows.

To the farmers, this system delivers a vastly accelerated damage verification and indemnification process. Because the platform utilizes meteorological indices to automate assessments rather than relying entirely on lengthy manual field inspections, farmers will experience much quicker insurance processing and payouts following extreme climate events. Rapid access to these funds helps mitigate their financial vulnerabilities and promotes faster recovery for the regional agricultural economy.

To the regional and national government, this study drives the digital transformation of public agricultural services while ensuring that all sensitive records are kept strictly secure.

By utilizing a secure, localized spatial database rather than vulnerable public clouds, it proves that government offices can successfully modernize disaster response, strengthen climate resilience in areas like Northern Mindanao, and improve service delivery without compromising the security and confidentiality of sensitive citizen information.

To the future researchers and developers, this project serves as a valuable technological baseline for software developers and academics interested in agricultural geoinformatics. It demonstrates a working, cost-effective framework for building secure, localized spatial platforms integrated with real-time weather data. This provides a practical foundation for future studies aiming to design innovative, non-cloud-based disaster response tools for government agencies in developing regions.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1. Preliminary Literature Review

The transition towards digital governance in agriculture relies heavily on optimizing spatial data management and mitigating environmental risks. This review examines the fundamental concepts, historical evolution, and current trends surrounding Geographic Information Systems, data centralization, and parametric agricultural insurance. The objective of this review is to synthesize existing local and international literature to establish the technical and theoretical foundations for developing a secure, localized WebGIS platform. By critically analyzing prior methodological approaches and identifying current research gaps in data privacy and disaster assessment, this review justifies the necessity of the proposed AgriSureGIS platform for the Philippine Crop Insurance Corporation Region X.

2.1.1. WebGIS and Spatial Decision Support in Agriculture

This section highlights the fundamental shift from traditional, standalone GIS to web-based platforms (WebGIS) within the agricultural sector.

The integration of GIS and internet technologies has fundamentally shifted agricultural data management towards web-based platforms. According to Patel et al. (2023), adopting open-source software such as Apache server, GeoServer, OpenLayers, and PostgreSQL with PostGIS extensions enables the development of interactive, dynamic platforms for agricultural monitoring and management. This web-based architecture allows users and decision-makers to access, visualize, and analyze spatio-temporal data over the internet without relying on monolithic, proprietary desktop software.

Handling the complex spatial and attribute data required for large-scale agricultural monitoring necessitates highly scalable system designs. To address this, Hu et al. (2024) proposed the construction of a digital agricultural GIS application suite utilizing microservice architecture and web componentization. This modular approach decomposes the system into independent service modules, allowing it to efficiently handle high concurrency and dynamic scaling while seamlessly integrating spatial data management, spatial analysis, and visualization.

Furthermore, transitioning to a WebGIS infrastructure facilitates simultaneous, multi-user collaboration, which is a significant advantage over traditional desktop GIS. Modica et al. (2025) highlighted that web-based multicriteria spatial decision support systems (MC-SDSS) allow multiple experts from various fields to interact with and evaluate geospatial data concurrently. Similarly, Avanidou et al. (2023) demonstrated the practical application of Web-GIS systems for remote farm management monitoring. By utilizing a central geodatabase linked to a web map application, their system enables land managers to remotely record, track, and analyze agricultural practices across numerous fragmented fields.

2.1.2. Eliminating Data Silos and Building Centralized Systems

This section highlights the critical challenges posed by fragmented data management, commonly referred to as “data silos”, and the operational benefits of transitioning toward centralized information architectures.

Data silos occur when information is isolated within different departments or stored in incompatible systems, which inherently hinders the free flow of information across an organization. According to Abu-AlSondos (2025), the persistence of these silos in the insurance sector causes significant delays, data inconsistencies, and suboptimal decision-making. Because different departments often rely on fragmented platforms and independent databases, organizations are forced to depend on manual data reconciliation, which is a time-consuming and error-prone process that inflates operational expenses. Furthermore, Abu-AlSondos (2025) emphasized that in highly regulated environments, the inability to generate clean, consolidated, cross-departmental data makes it practically impossible to meet strict reporting standards, thereby exposing insurers to severe compliance penalties and reputational risks.

Addressing these operational inefficiencies requires a strategic shift toward centralized computing resources and data management. Gordon (2020) highlighted that the centralization of Information and Communication Technology (ICT) offers significant advantages to organizations, including economies of scale, standardized equipment and processes, and a reduction in management overhead. Centralizing an organization's information systems ensures consistency in the data collected and significantly improves the speed and ease of aggregating data for crucial decision-making. While Gordon (2020)

acknowledged the necessity of decentralized access to maintain user productivity, the core architecture must remain centralized to effectively rationalize control and secure continuous operations.

In the context of geospatial and environmental data, eliminating silos is most effectively achieved through the implementation of Spatial Data Infrastructures (SDI). Wetzel et al. (2024) demonstrated that establishing an SDI with standardized WebGIS servers, such as GeoServer, is crucial for harmonizing fragmented data and making it interoperable across different platforms. By utilizing centralized geospatial databases and open-source web services, organizations can replace isolated data storage such as standalone desktop files with a unified system that supports seamless data exchange and concurrent multi-user access. This infrastructure allows various stakeholders to simultaneously access, search, and utilize the exact same updated spatial information, effectively dismantling previous data barriers.

2.1.3. Multi-Criteria Filtering for Spatial and Attribute Data

Effective spatial decision-making in large-scale agricultural systems depends on the ability of users to rapidly query and isolate specific data points to prevent user fatigue when browsing through thousands of records .

Cui et al. (2022) emphasize that modern WebGIS platforms must incorporate advanced filtering tools that allow users to isolate parcels based on unique features, such as land use type, field code, or individual producer IDs . For example, the FarmGeoBalance system described by Avaniidou et al. (2023) utilizes a Search and Select service that enables managers to group and view agricultural practices across numerous fragmented fields through a unified interface .

This interaction allows a Click-Feature event to select a plot and automatically highlight it on the map while simultaneously displaying its corresponding metadata such as area, coordinates, and planting date in a structured attribute box . Such multi-criteria tools are essential for GIS specialists to generate Payout-Ready summaries for management review without manually searching through voluminous datasets

2.1.4. Integration of Meteorological Data for Automated Risk Assessment

This section highlights the importance of integrating real-time meteorological data into spatial platforms to transition from manual claims verification to automated disaster risk and agricultural damage assessments.

The integration of meteorological data into spatial platforms has become a cornerstone of modern agricultural insurance and disaster risk management. Traditional claims verification processes are highly tedious and resource-intensive, but utilizing weather data models can significantly reduce these administrative burdens. According to Nguyen et al. (2025), integrating climate and weather data helps reduce basic risk and improves the design of agricultural index-based insurance. Similarly, Nguyen-Huy et al. (2024) emphasized that weather and rainfall index-based insurance can effectively mitigate financial risks for farmers exposed to extreme climate events. By integrating these meteorological variables into geoinformatic systems, agencies can rapidly estimate damages and streamline the payout process for vulnerable small-scale farmers (Rabuh et al., 2024). Furthermore, providing tailored regional climate information through Spatial Data Infrastructures (SDI) ensures that local authorities can implement evidence-based disaster responses efficiently (Wetzel et al., 2024). AgriSureGIS adopts this principle by introducing a module that automatically extracts spatial hazard data from PAGASA Tropical Cyclone Bulletins and overlays it against the PCIC Risk Exposure List. This approach not only streamlines the damage assessment workflow during typhoon events but also promotes the overall resilience of the agricultural economy (Chen et al., 2025)

2.1.5. Automated Meteorological Data Acquisition and Notification

Transitioning from manual monitoring to an automated disaster response framework requires internal scripts capable of real-time data pulling from meteorological bureaus. Wetzel et al. (2024) describe a robust implementation using Python-based scripts that regularly iterate over external data folders or websites to automatically detect and upload new contents to a WebGIS server via an API.

To maximize responsiveness, these automated workflows include notification modules such as on-screen pop-ups or automated email alerts that trigger immediately when a new report is parsed and available. This automation ensures that disaster risk assessments, which

were previously delayed by the manual downloading of files into disconnected repositories, can proceed instantly as new weather data becomes available . This streamlined process is critical for achieving the project objective of providing insurances to farmers within three to five days after a typhoon exits the Philippine Area of Responsibility (PAR).

2.1.6. Secure Localized Spatial Databases for Information Management

This section highlights the critical role of secure, localized spatial databases in centralizing large-scale agricultural records while ensuring strict data privacy.

Transitioning from fragmented data files such as standalone CSV and GPX files to a centralized spatial database is critical for handling large-scale agricultural records efficiently. According to Abu-ElSondos (2025), eliminating data silos is essential in the insurance sector, as fragmented information restricts strategic analytics and leads to delays, compliance failures, and poor decision-making. Modern spatial databases function as robust engines for geographical analysis. Specifically, object-relational database management systems like PostgreSQL, combined with its spatial extension, PostGIS, provide highly advanced features for storing, querying, and managing massive spatial datasets (Hajisheko et al., 2025; Ilba, 2021). Furthermore, utilizing localized database architectures addresses the growing necessity for strict information privacy. Shahriar et al. (2025) noted that processing large volumes of personal information poses unique privacy risks, making robust privacy-enhancing solutions imperative. By implementing a secure, localized PostGIS database instead of a public cloud repository, government agencies like the PCIC can establish strict access controls. This localized infrastructure effectively protects sensitive farmer data in compliance with data privacy regulations while simultaneously overcoming operational bottlenecks (Tissayakorn, 2025).

2.1.7. Hybrid Deployment Progressive Web Applications (PWA) and Native Frameworks

The evolution of Geographic Information Systems (GIS) deployment now favors hybrid models that ensure system resilience across varying network conditions. Ginzburg and Padalko (2020) propose the implementation of Progressive Web Applications (PWA) for GIS

interfaces, highlighting their ability to provide fault-tolerant data delivery and complete offline operation on any stationary or mobile device equipped with an HTML5-compliant browser.

This architecture utilizes advanced caching mechanisms to allow specialists to work with previously loaded spatial data even when the server is disconnected, featuring automatic synchronization of user inputs once connectivity is restored. A significant institutional benefit is the reduction in network traffic; research indicates that PWAs can save over 80% in monthly data transfer compared to traditional web applications, which is vital for regional offices with limited bandwidth.

In institutional environments like the PCIC, these web-based tools are often paired with a native "Desktop terminal" subsystem, as seen in the research by Yining et al. (2025), where a native environment is dedicated to intensive pre-field configuration, lab-based data integration, and high-level quality control before broader deployment.

2.1.8. A WebGIS Ecosystems Utilizing Google Earth Engine

Several established research frameworks demonstrate the efficacy of integrating Google Earth Engine as a backend analytical engine for WebGIS applications. Bhakal et al. (2024) developed a specialized model within the GEE framework that enables users to visualize and analyze geospatial data for a wide range of precision agriculture applications, such as assessing crop health through various indices.

In the context of disaster management, Thakur et al. (2026) utilized GEE to create an innovative framework for rapid impact assessment using Sentinel-1 Synthetic Aperture Radar data. Their system leverages the GEE API to deliver workflows that provide preliminary damage assessments within hours of new data acquisition, bypassing time-consuming manual analysis.

This integration is particularly valuable for parametric insurance because GEE can process massive multi-temporal datasets to verify planted areas even during typhoon events by utilizing SAR imagery that penetrates thick cloud cover, which typically obstructs traditional optical sensors.

2.1.9. Synthesis of the Reviewed Literature

Despite these advancements, a distinct research gap exists in current literature and system development. The majority of recent agricultural WebGIS frameworks heavily emphasize either public cloud integration or mobile-offline field collection applications. There is a lack of literature addressing the specific institutional needs of government agencies that require strictly localized, non-cloud spatial databases PostGIS to comply with stringent data privacy laws, coupled with automated parametric assessment tools tailored to local meteorological bureaus.

The proposed AgriSureGIS system directly fills this gap. By utilizing an open-source FOSS architecture, it establishes a highly secure, localized platform that eliminates the need for mobile-cloud dependencies. Concurrently, by developing a Parametric Insurance Indemnification Assessment Tool that cross-references PAGASA typhoon bulletins against farm polygons, this study extends the existing knowledge of spatial decision support systems, offering a tailored, modernized workflow for disaster response in PCIC Region X.

2.2.10 Comparison of Existing Systems/Studies

To establish the foundation for this IT Capstone Project, a review of existing studies was conducted. Table 1 presents a comparative analysis of related works, highlighting key methodologies, system features, technological frameworks, and research gaps. This comparison provides insights into how previous studies approached similar problems and helps identify areas for improvement and innovation in the proposed system.

Feature	AgriSureGIS	CONNECTOSED	ReKIS	WebGIS-based decision support system	WebGIS Suitability Olive	Sustainable GeoInformatic Insurance	GIS-based intelligent system for geochemical sampling	WebGIS-based platform for effective landslide hazard
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								manage ment
Spatial Data Infrastructu re with OGC-Comp liant Web services	✓	✓	✓	✓	✓	✓		✓
Spatial-to- Attribute Data Linking	✓					✓	✓	✓
Parametric Insurance & Meteorolog ical Integration	✓					✓		
Automated Risk and Damage Assessment Module	✓	✓	✓	✓	✓	✓		✓
Advanced Location Search & Filter	✓		✓	✓				✓
Secure User Access Levels	✓		✓		✓		✓	✓
Interactive Visual Dashboards	✓	✓	✓	✓	✓			✓

Table 1. Comparative Analysis of Related Works

Gantt Chart

The IT Capstone Project follows a structured timeline, ensuring the completion of key milestones from February to October 2026. The project is divided into three main phases:

1. *Concept Presentation Defense (February 2026)*: Focuses on writing Chapters 1-2 and conducting initial research.
2. *Project Proposal Defense (May 2026)*: Requires the completion of the system's wireframe and revisions of Chapters 1-3.
3. *Final Defense (October 2026)*: Involves system development, testing, and documentation (Chapters 4-5).

The Gantt Chart (Figure 1) illustrates the detailed schedule and dependencies among tasks.

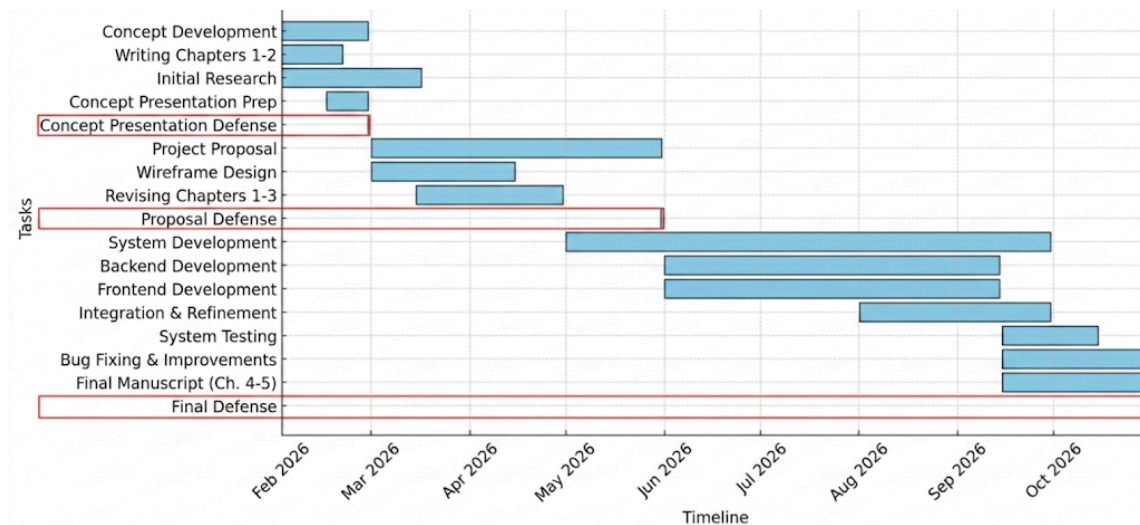


Figure 1. Gantt Chart

CHAPTER III

METHODOLOGY

3.1. Software Development Model

The development of AgriSureGIS adopts the Agile Development Model, a methodology selected for its flexibility in accommodating evolving requirements through iterative and incremental cycles. This approach enables complex functionalities such as real-time PAGASA bulletin extraction and PostGIS spatial data integration to be continuously refined and optimized across successive technical sprints, rather than relying on a single deployment phase.

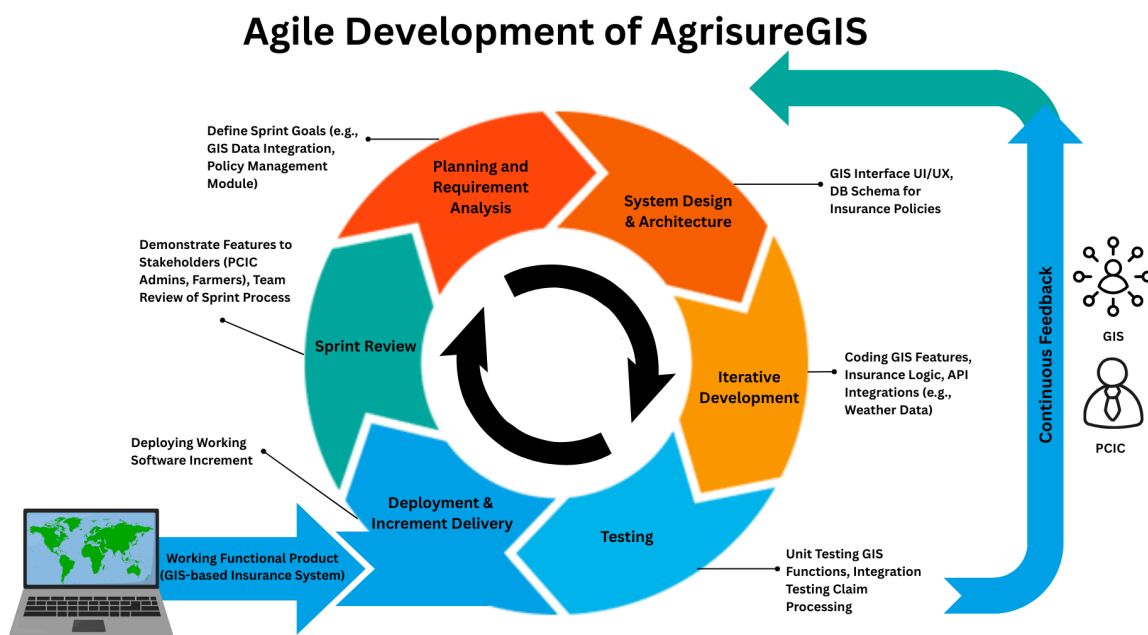


Figure 2. Agile Development Model

For the development of the AgriSureGIS platform, the proponents selected the Agile Software Development Life Cycle (SDLC), specifically utilizing a framework structured around iterative development cycles. This methodology was deliberately chosen over rigid, sequential models (such as the Waterfall model) to address the specific operational and technical realities of building a bespoke system for a government agency.

During the initial requirements elicitation phase, it became evident that the PCIC Region X operates in a dynamic environment where reporting standards, UI preferences, and assessment rubrics may require refinement. The Agile methodology embraces changing requirements, even late in development. By breaking the project into iterative cycles, the development team can seamlessly integrate new client requests without derailing the entire project timeline.

A core principle of this project is ensuring that the final system perfectly mirrors the actual workflow of the PCIC GIS Specialists. The Agile model mandates a "Sprint Review" at the end of each cycle. This continuous feedback loop guarantees that the system is built with the client rather than just for them, effectively eliminating the risk of delivering an end-product that the agency cannot use. Moreover, the system relies heavily on advanced external integrations, specifically the Python-based scraping of PAGASA Tropical Cyclone Bulletins and the spatial radar processing via the GEE API. These are high-risk technical components. The Agile framework allows the team to isolate these complex modules into dedicated sprints.

3.2. Planning and Requirement Analysis

3.2.1 Information Gathering

During this phase, the proponents engaged directly with the stakeholders of the Philippine Crop Insurance Corporation (PCIC) Region X to define the exact system boundaries, identify operational bottlenecks, and establish the technical feasibility of the proposed WebGIS solution. The team conducted a series of three comprehensive, on-site interviews at the PCIC regional office. These sessions were deliberately scheduled around the client's availability to ensure maximum participation from the primary end-users, specifically the GIS Specialists and management evaluators. The proponents also utilized synchronous virtual meetings via Google Meet ensuring that rapid clarifications could be made without disrupting the agency's daily operations. To identify the exact operational "pain points" in real-time, the proponents conducted direct observations of the agency's current procedures. PCIC management provided a controlled, discreet demonstration of their manual disaster risk assessment workflow, including an overview of the legacy PCIC Automated Business System

(PABS). During this demonstration, strict data privacy protocols were enforced; all sensitive information regarding the farmers was redacted or hidden from the proponents.

Data gathering was highly iterative. Following each interview, the development team conducted internal deliberations to dissect the gathered data. The proponents mapped out the agency's existing manual disaster risk assessment workflow specifically the manual downloading of PAGASA Tropical Cyclone Bulletins and the standalone QGIS tracing and evaluated how these processes could be translated into programmable algorithms.

The actual process starts with the Farmer registration where the farmers can either call or go to the office to request for an Insurance Application Form. Farmers who are registered under the RSBSA (Registry System for Basic Sectors in Agriculture) receive free government insurance. One (1) month after the insurance application, a post-planting inspection is conducted (if renewal), and an on-site georeferenced polygon of the farmer's land is created.

- The insurance coverage and the mapped polygons are strictly focused on rice areas for the meantime. Although rice and corn can be insured.
- **Requirement:** It is a requirement that all insured farmers have an equivalent polygon.

The GIS specialist manually performs two separate tasks:

First, Georeferencing is a process of using location-based technologies (GPS, GIS, drones, and satellite images) to:

- Map exact field boundaries of insured rice plots.
- Verify claims based on spatial evidence.
- Monitor crop conditions and damage remotely.
- Streamline insurance processing via PCIC Automated Business System Georeferencing (PABS-Georeferencing).
- **Hardware/Tools Used:** Gadget/tablet using GPS to pinpoint the exact hectare.
- **Device (GPS):** receives signal from satellite, uses triangulation method, enables location, then allows creation of a polygon application.

In which Necessary Data are gathered for the assessment:

- Date of planting
- Harvest area (hectares)

- Harvest boundaries
- Farm polygon

Second, The main Legacy PCIC Automated Business System where they store, compile and export all their profiling and data:

- Profiling Data Elements:
 - Farmer Name
 - Age
 - Sex
 - Address
 - Spouses
 - Beneficiary
 - Farm Area (Hectares)
 - Farm ID
 - Farm Address
 - Boundaries (Names of Neighbors: Example 1, Farmer 2)
 - Amount of Cost (Regular)

The said process includes the farmer filling out an application form wherein they declare their farm area. Then, 1 month or 2 weeks after, a post-planting inspection happens for PCIC to verify every detail the farmer listed and if it is accurate. It is the duty of the PCIC adjuster/validator to correct and verify if the declared area is accurate.

- **Data Storage:** The established polygons are stored in the PCIC database, specifically on their local servers.

Georeferenced files are the critical component that allows PCIC to process payouts without needing to physically visit the farm after a disaster.

- **Timeline Constraint:** A typical cropping cycle lasts for 3 months. The insurance coverage is directly tied to this timeline. Insurance expires after the harvest, meaning farmers must renew their insurance for every single planting cycle.

The next manual process involves typhoon monitoring. When a typhoon enters the Philippine Area of Responsibility (PAR), PAGASA uploads an initial bulletin report with a 1-hour delay. PAGASA continues to upload a newly updated bulletin every 3 hours.

- Bulletin reports are released strictly as PDF files. PAGASA has not updated the template of these reports, meaning the format remains consistent over time.
- PAGASA will then remove the PDF files 1–2 weeks after the typhoon.

This is where the GIS Specialist manually downloads the PAGASA bulletin PDF from the website. After downloading, the analyst manually uploads and stores these PDF files into Microsoft OneDrive. Right after, The analyst opens the PDFs and manually extracts the necessary typhoon details and inputs this data into a CSV file extracting only the necessary data needed:

- Time/Date
- TCWS No. (Tropical Cyclone Wind Signal) - has a fixed wind velocity range:
 - Wind Signal no. 2 = 62-88km/h, Wind Signal no. 3 = 89-117km/h, Wind Signal no. 4 = 118-184km/h, Wind Signal no. 5 = 185km/h+
- Affected Areas

Wind signal no. 2-5 are the only ranges where the affected farmers are eligible for the indemnification process.

Then, the specialist loads the necessary data into QGIS:

- The manually compiled CSV file is then loaded into QGIS software.
- **Primary Purpose:** To visualize the data.
- The analyst will then manually input and trace the storm's trajectory based on the CSV data.

Through this manual tracing, the analyst determines which measures exactly how many hours a specific municipality was exposed to the typhoon signal.

Critical Focus: The risk assessment explicitly focuses on the damage afflicted by strong winds (wind velocity) rather than flooding.

3.2.2 Software Requirements Specifications

A. Functional Requirements

- The specialist shall have an account in the system.

- The specialist shall have access to all functionalities in the system.
- The system shall allow the specialist to import the existing risk exposure report (farmer records/profiles) via CSV file.
- The system shall allow the specialist to upload the farmers' spatial GPX files to establish farm polygons.
- The system shall allow the specialist to execute the automated analysis, triggering the system to generate the Tropical Cyclone Bulletin (TCB) data, which includes period of exposure and wind velocity.
- The system shall automate the calculation process by mapping the period of exposure, identifying the wind velocity, matching the crop growth stage, and calculating the corresponding indemnity/damage factor.
- The system shall allow the specialist to filter the dashboard and reports by municipality, province, barangay, or city.
- The system shall allow the specialist to sort records alphabetically, by planting date, or by any specific column.
- The system shall allow the specialist to generate and export an extended CSV report that appends the calculated results (period of exposure, wind velocity, indemnity factor, and final indemnity payment) to the original PCIC row format.
- The system shall allow the administrator to securely log in to access the backend infrastructure.
- The system shall allow the administrator to create or restrict user accounts, manage localized system configurations, and execute database backups to ensure institutional data security.
- The system shall provide a dedicated "Calibration/Settings" module on the dashboard for configuring system parameters.
- The system shall allow the administrator to input or manage Google Cloud Project credentials to allocate the capacity for Google Earth Engine SAR analysis.
- The system's Python-based parser shall monitor the PAGASA website and automatically download the Tropical Cyclone Bulletin updates during an active typhoon in the PAR.
- The system shall trigger an on-screen pop-up and an automated email alert to notify the GIS specialist when a new bulletin is successfully parsed.

- The system shall automatically compile and store all downloaded versions of the PAGASA PDF bulletins during a typhoon event without overwriting previous versions.
- The system shall interface with Google Earth Engine to retrieve Synthetic Aperture Radar (SAR) imagery to analyze if the area is planted within the polygons.
- The main dashboard shall be structurally divided into four specific modules to handle the automated disaster risk assessment workflow: a Home Dashboard, Spatial Analysis & Data Import, Assessment & Reporting, and Calibration/Settings.
- The system shall provide a drag-and-drop area for uploading new CSV files.
- The system shall display corresponding metadata (farmer ID, name, address, farm ID, hectare area, and whether it is planted) instantly when a user clicks on a farm polygon on the map.
- The system shall display warning prompts before a user deletes, overwrites, or finalizes any data.
- The system shall provide a "Generate Report" button that allows users to preview the CSV layout before finalizing the download
- The system shall utilize specific UI theme colors combining green, yellow (gold), and blue, and support both light and dark modes.

B. Non-Functional Requirements

- **Accessibility** - The web interface shall be dynamically responsive so that it is readable and functional on varying screen sizes, including smaller monitors used during field office meetings.
- **Accuracy** - The system must accurately extract variables such as time, Tropical Cyclone Wind Signal number, wind velocity, and affected areas directly from PAGASA's PDF bulletins.
- **Compatibility** - The system shall be compatible with and accessible across multiple web browsers.
- **Performance** - The system shall run heavy geoprocessing calculations in the background while displaying a loading spinner to the user.

- **Usability** - The graphical interface must be simple, functional, and easy to navigate, integrating PCIC's official theme colors of green, yellow (gold), and blue.
- The interface must support both light and dark modes.
- The system must prompt the user with warning messages before allowing them to delete, overwrite, or finalize data.
- **Reliability** - The system must automatically save ongoing work and imported data.
- **Security** - Spatial boundaries and sensitive insurance data must be securely hosted on a localized PostgreSQL database with the PostGIS extension, specifically avoiding public cloud servers.
- The system must retain the last active session to prevent data loss, in the event of a session timeout due to 5 minutes of inactivity.
- **Data Consistency** - The system must generate consistent CSV files that the PCIC finance department can utilize without manual editing, appending the calculation columns to the end of the existing PCIC format.
- **Fault Tolerance** - If the system fails to parse the PAGASA PDF bulletins, it must explicitly alert the user with specific error messages rather than crashing.
- The system must be designed to retain baseline functionality even if the PAGASA website goes down or experiences an outage.

C. System Modules

- **Monitoring and Extraction Module:** This module serves as the landing page and active monitoring center for the system. It handles the real-time pulling and extraction of PAGASA Tropical Cyclone Bulletin reports and features a notification system to immediately alert the GIS specialist when a new TCB is successfully parsed and downloaded.
- **Spatial Analysis and Data Import Module:** This module maximizes visualization and data handling by dividing the interface into a top and bottom layout. The top panel features the interactive GIS base map that displays the overlay of the typhoon's trajectory and the mapped polygons of the affected insured rice farms, while also allowing the export of the initial "Period of

Exposure" report. The bottom panel provides a drag-and-drop interface for importing the legacy farmer record CSV files and spatial GPX polygons, displaying a tabular overview where clicking a row automatically highlights the corresponding farm polygon on the map above.

- **Assessment and Reporting Module:** This module serves as the data compilation and review center for indemnification processing. It presents a summarized view of the parsed TCB records alongside the imported rice risk exposure records, automates the extended tabular calculation results (factoring in the period of exposure, wind velocity, crop growth stage, and indemnity factors), and features a "Generate Report" button to preview and export the finalized CSV indemnification report.
- **Calibration and Settings Module:** This module functions as the administrative and configuration panel, accessed via a toggle menu to maximize the vertical screen space for the map interface. It allows authorized users to manually edit system parameters or override data as a fail-safe and manage system credentials such as the Google Cloud Project IDs used to allocate capacity for Google Earth Engine satellite analysis.

3.3. System Design

3.3.1 System Architecture

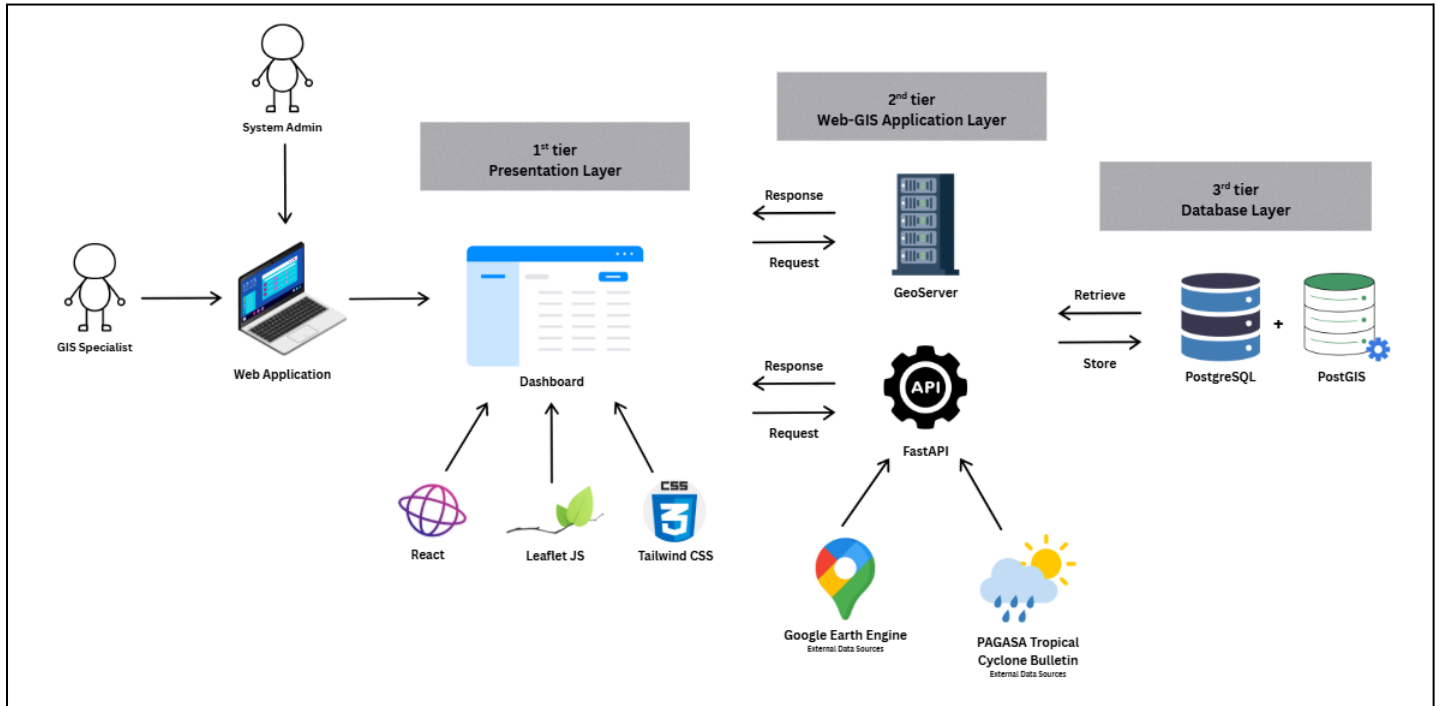


Figure 3. System Architecture

As shown in Figure 3, the proposed system utilizes a robust three-tier architecture tailored for spatial data processing and web-based visualization. The first tier, or Presentation Layer, provides the user interface for the GIS Specialist and System Administrator through a dynamic web dashboard built with React, styled with Tailwind CSS, and integrated with Leaflet JS for interactive map rendering. The second tier, the Web-GIS Application Layer, acts as the system's processing core; it employs FastAPI to manage API routing and fetch external meteorological and satellite data from the PAGASA Tropical Cyclone Bulletin and Google Earth Engine, while GeoServer operates as the dedicated spatial engine to process and serve geographic data. Finally, the third tier, the Database Layer, ensures secure and persistent storage by leveraging PostgreSQL for structured relational data alongside the PostGIS extension for advanced spatial querying and geospatial data management.

3.3.2 Use Case Diagram

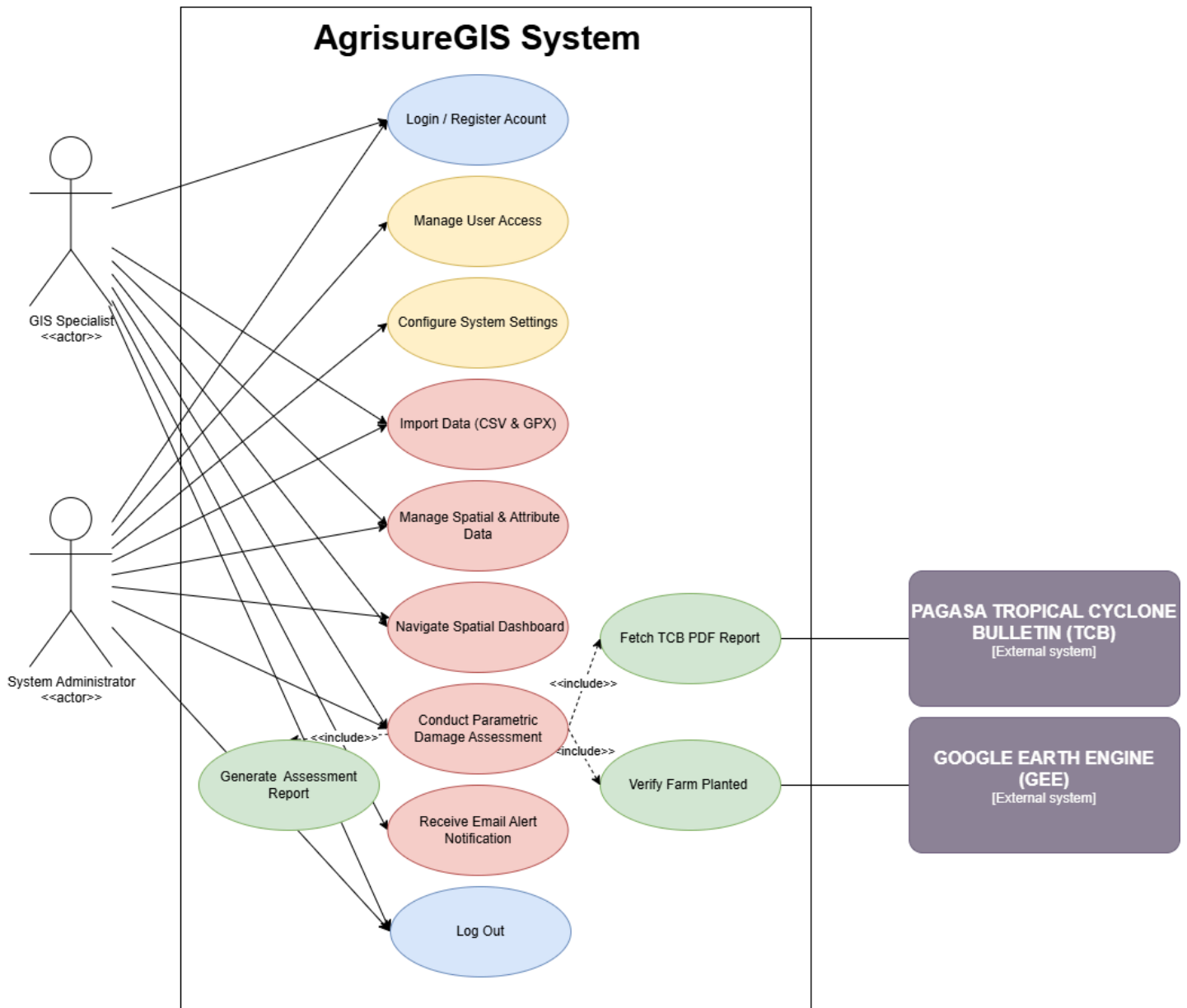


Figure 4. Use Case Diagram

The system's use case is shown in figure 5, illustrating the interactions between two primary human actors and two external system entities required to execute the agricultural risk assessment workflows. The internal actors include the GIS Specialist, who serves as the primary operational user tasked with managing spatial and attribute data including the importation of Risk Exposure Lists and GPX files, navigating the spatial dashboard, conducting parametric damage assessments, receiving email alert notifications and generating final reports; and the System Administrator, who oversees system governance, account

authentication, and user access. To support these core functions, the system integrates with secondary external entities. Specifically, it employs an include relationship with the PAGASA Tropical Cyclone Bulletin to fetch essential meteorological reports during an assessment, and a relationship with Google Earth Engine to conditionally verify planted farm statuses using remote sensing data.

3.3.3 Context Diagram

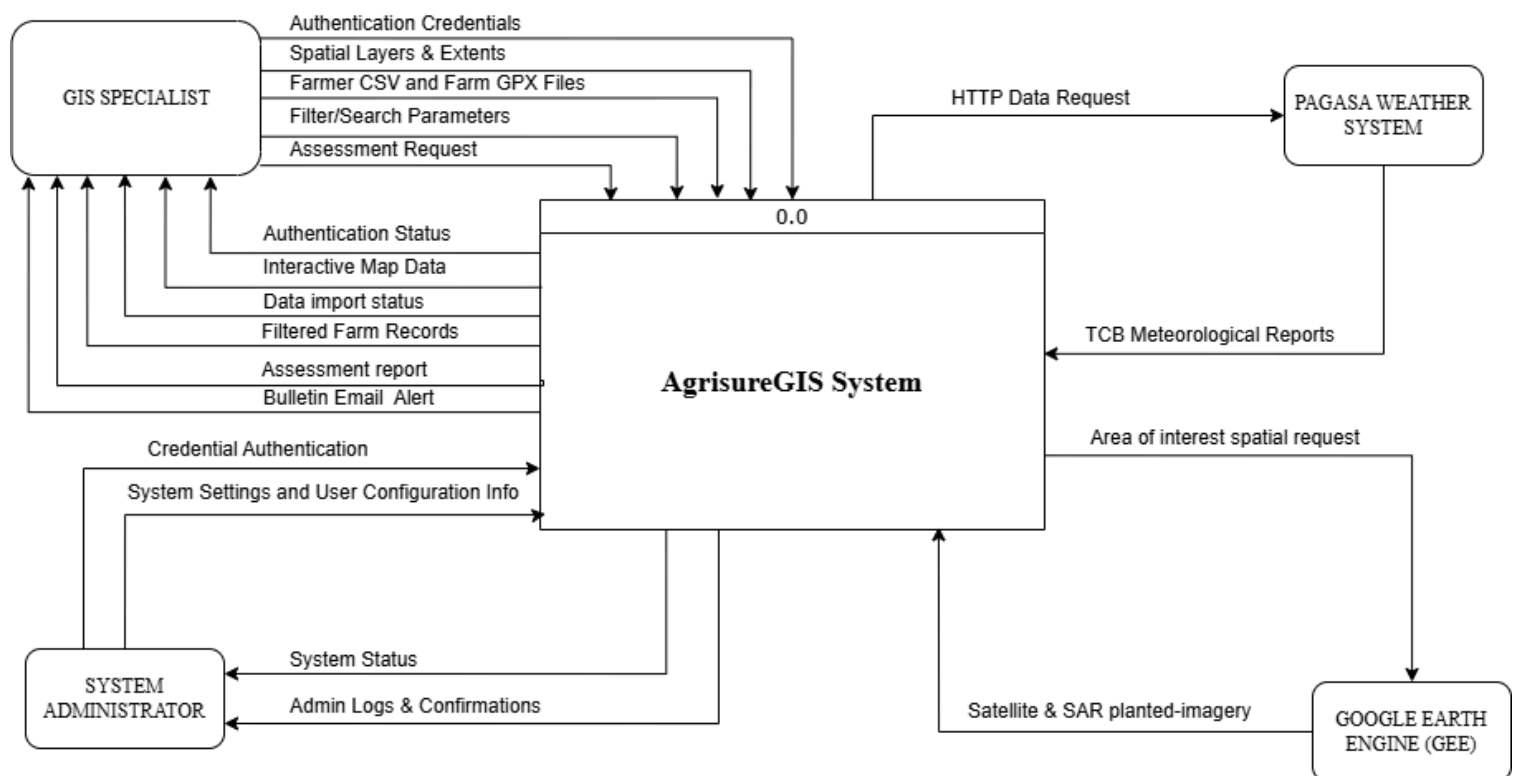


Figure 5. Context Diagram

At the operational core shown in Figure 6, is the GIS Specialist, who supplies the system with critical inputs by entering authentication credentials, requesting spatial layers and extents, uploading farmer CSV and farm GPX files, applying filter and search parameters, and initiating assessment requests. In response, the system processes these interactions and outputs the authentication status, interactive map data, data import status, filtered farm records, exported assessment reports, and bulletin email alerts directly back to the user. System governance is maintained by the System Administrator, who interacts with the central system by entering authentication credentials and configuring user and system settings. The system subsequently returns the overall system status, alongside administrative logs and

confirmations, to this actor. To execute its specialized spatial and parametric analyses, the AgrisureGIS System automatically interfaces with secondary external entities. It actively sends requests for TCB bulletins to the PAGASA Weather System, which in turn provides the necessary TCB bulletin reports. Concurrently, the system submits Area of Interest (AOI) spatial queries to Google Earth Engine, which responds by returning crucial satellite imagery and planted-status data.

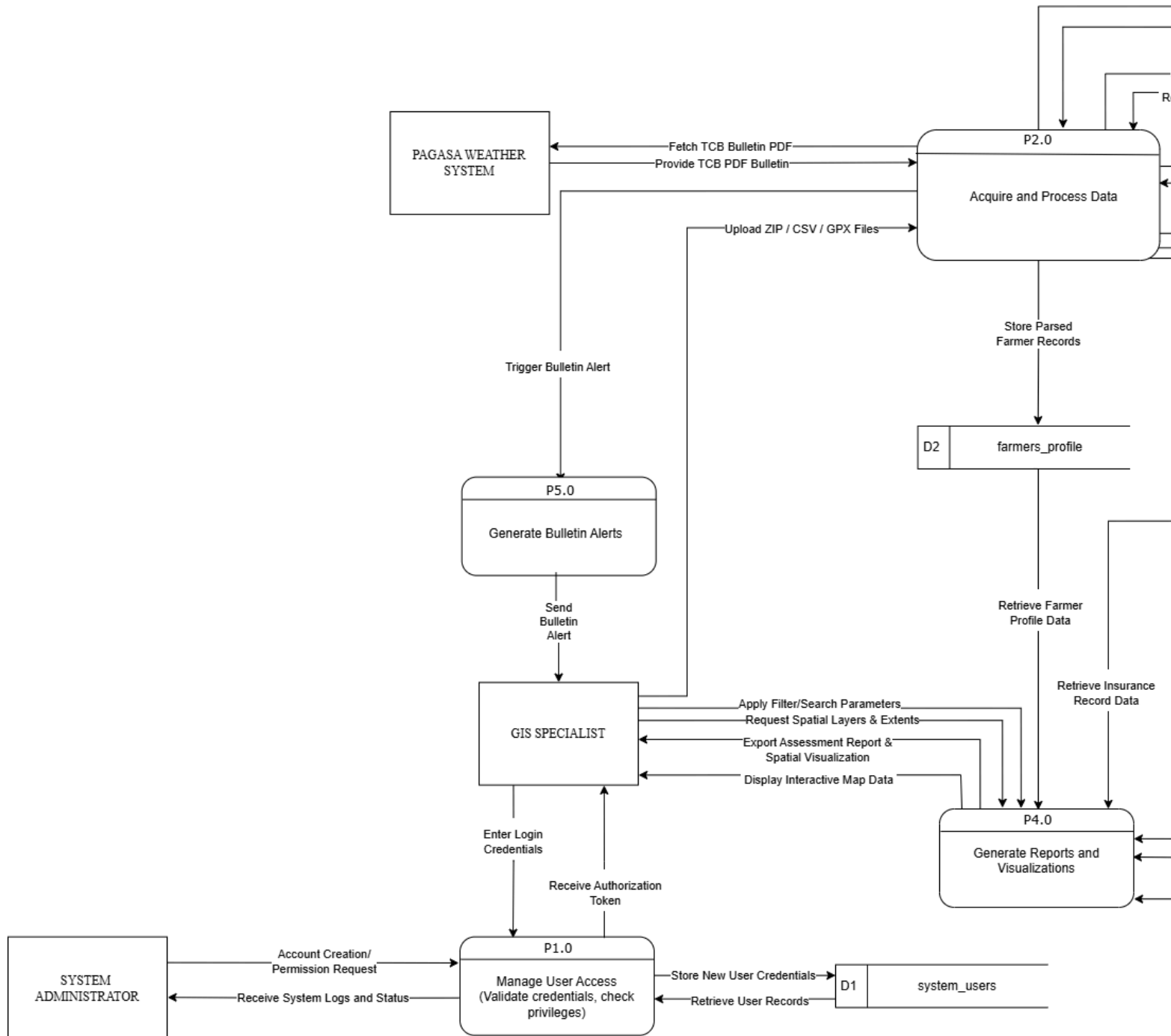


Figure 6.1 Data Flow Diagram left side

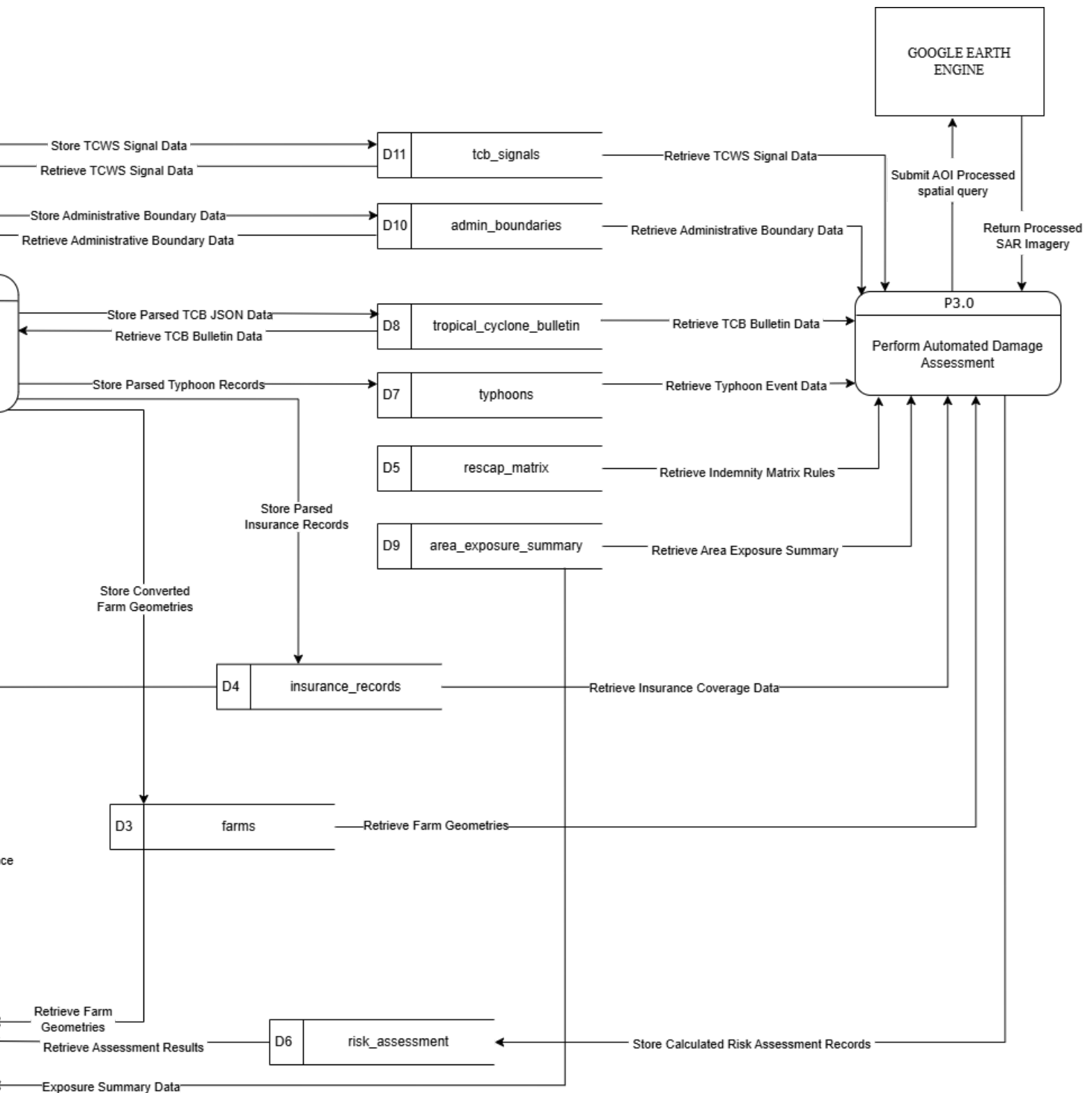


Figure 6.2 Data Flow Diagram right side

The Level 1 Data Flow Diagram is shown in Figure 6, illustrating how the AgriSureGIS system decomposes its main operations into five functional processes that coordinate data movement between the system's actors, external entities, and internal data stores. The diagram presents the flow of information involving two primary human actors, namely the GIS Specialist and the System Administrator, and two external system entities, namely the PAGASA Weather System and Google Earth Engine. These entities support the system's agricultural risk assessment workflow by providing user inputs, administrative access control, meteorological data, and satellite-based verification data.

Process 1.0, Manage User Access, handles account authentication and access control for the GIS Specialist and System Administrator. This process receives login credentials and account or permission requests, validates user records, checks user privileges, and stores or retrieves credentials from the `system_users` data store (D1). It returns authorization tokens, system logs, and status information to ensure that only authorized users can access the system and perform their assigned functions.

Process 2.0, Acquire and Process Data, manages the system's data acquisition and ETL operations. This process receives ZIP, CSV, and GPX file uploads from the GIS Specialist, extracts and validates the submitted files, parses farmer and insurance records, and converts GPX-based farm boundaries into GeoJSON format. The processed farmer records, farm geometries, and insurance records are stored in `farmers_profile` (D2), `farms` (D3), and `insurance_records` (D4), respectively. The process also communicates with the PAGASA Weather System by requesting and receiving Tropical Cyclone Bulletin PDF data. The received bulletin is parsed into JSON format and transformed into structured typhoon, boundary, bulletin, and TCWS signal data, which are stored in `typhoons` (D7), `tropical_cyclone_bulletin` (D8), `admin_boundaries` (D10), and `tcb_signals` (D11).

Process 3.0, Perform Automated Damage Assessment, uses the processed agricultural, insurance, meteorological, and spatial data to compute risk assessment results. It retrieves farm geometries, insurance coverage data, typhoon event data, Tropical Cyclone Bulletin data, TCWS signal data, administrative boundary data, area exposure summaries, and indemnity matrix rules from the corresponding data stores. The process also submits farm polygon or SAR analysis requests to Google Earth Engine and receives processed SAR assessment statistics for plant-status verification. Using these inputs, the system calculates the corresponding risk assessment and stores the computed results in `risk_assessment` (D6).

Process 4.0, Generate Reports and Visualizations, produces the user-facing outputs of the system. This process retrieves farmer profile data, farm geometries, insurance records, exposure summaries, and assessment results to display interactive map data, apply filtering and spatial layer requests, and generate assessment reports with spatial visualizations. The generated output may also be exported as a final CSV assessment report for documentation, review, and further processing.

Finally, Process 5.0, Generate Bulletin Alerts, supports the system's notification function. This process receives a bulletin alert trigger after a PAGASA bulletin has been successfully parsed and processed by the system. It then sends a bulletin alert to the GIS Specialist to notify the user that new weather and hazard data are available for visualization, verification, and automated assessment.

3.3.5 Entity Relationship Diagram

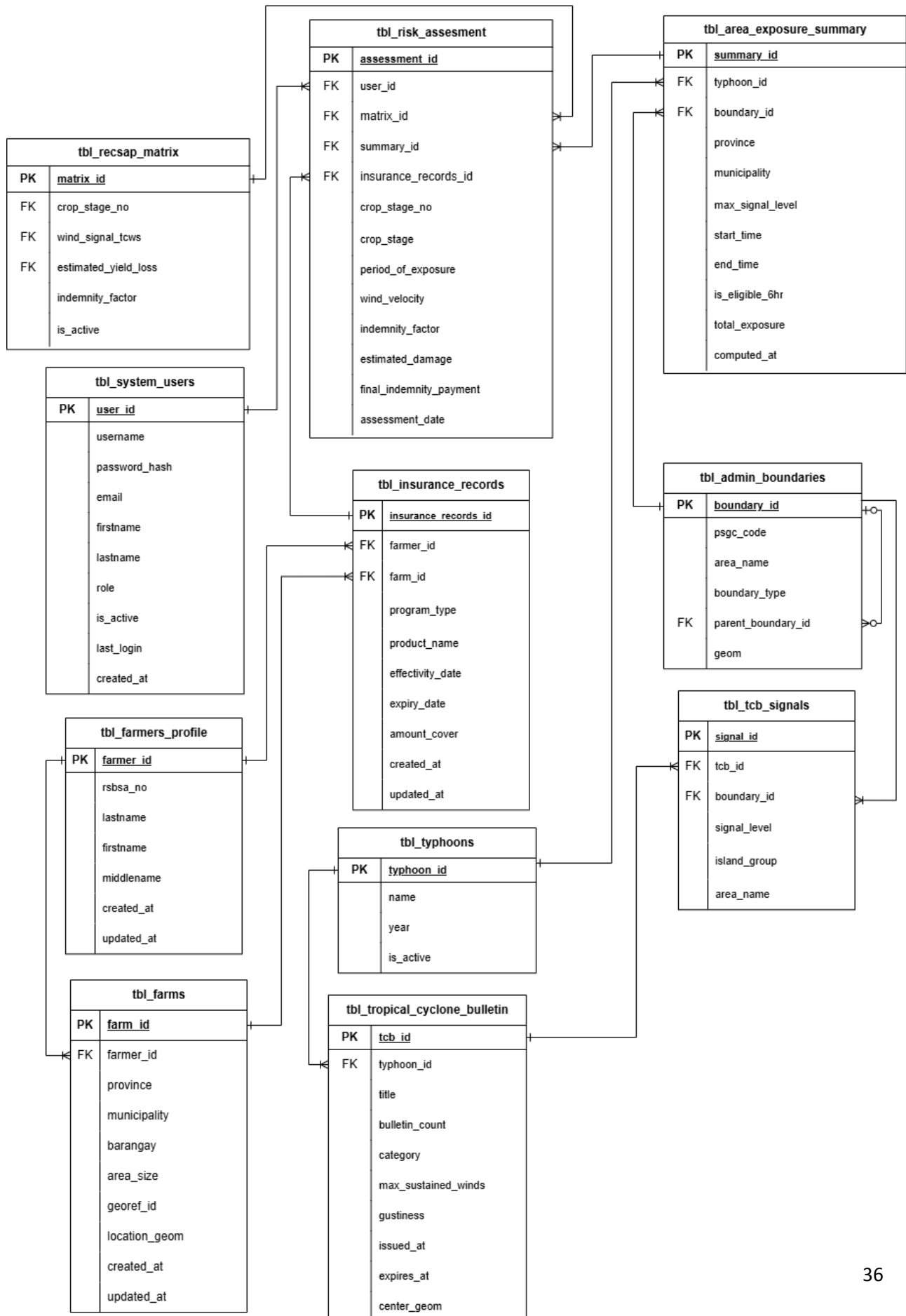


Figure 7. Entity Relationship Diagram

3.3.5.1 Database Dictionary

Table: tbl_system_users				
Field Name	Data Type	Length	Constraint	Description
user_id	SERIAL	50	AUTO_INCREMENT, PRIMARY KEY	Auto-incrementing unique ID for the staff member.
username	VARCHAR	50	UNIQUE, NOT NULL	The username used to log into the platform.
password_hash	VARCHAR	255	NOT NULL	The cryptographically hashed the password.
email	VARCHAR	100	UNIQUE, NOT NULL	Staff member's official email address.
firstname	VARCHAR	100	NOT NULL	Staff member's first name.
lastname	VARCHAR	100	NOT NULL	Staff member's last name.
role	ENUM	50	NOT NULL	Defines their permission level.
is_active	BOOLEAN	-	NOT NULL, DEFAULT TRUE	Toggle to easily suspend an account without deleting their data.
last_login	TIMESTAMPTZ	-	NULLABLE	Timestamp of the user's most recent successful login.
created_at	TIMESTAMPTZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAM P	When the account was created.
Table: tbl_farmers_profile				
Field Name	Data Type	Length	Constraint	Description
farmer_id	INT	50	PRIMARY KEY	Unique identifier for the farmer.
rsbsa_no	VARCHAR	50	UNIQUE, NULLABLE	Registry System for Basic Sectors in Agriculture number.

lastname	VARCHAR	100	NOT NULL	Farmer's last name.
firstname	VARCHAR	100	NOT NULL	Farmer's first name.
middlename	VARCHAR	100	NULLABLE	Farmer's middle name.
created_at	TIMESTAMPTZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the farmer profile was created.
updated_at	TIMESTAMPTZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the farmer profile was updated.
Table: tbl_farms				
Field Name	Data Type	Length	Constraint	Description
farm_id	INT	50	PRIMARY KEY	Unique identifier for the land parcel.
farmer_id	INT	50	FOREIGN KEY	References farmers.farmer_id. Connects land to the owner.
province	VARCHAR	100	NOT NULL	Regional location of the farm.
municipality	VARCHAR	100	NOT NULL	City or town location of the farm.
barangay	VARCHAR	100	NOT NULL	Local neighborhood/village of the farm.
area_size	DECIMAL	10, 4	NOT NULL	Size of the land in hectares or square meters.
georef_id	VARCHAR	100	NULLABLE	The specific map code used to link this record to your spatial files (e.g., "R10-13-16-012-000016").
location_geom	GEOMETRY	GEOMETRY(MultiPolygon, 4326)	NULLABLE	PostGIS spatial data representing the exact boundary of the farm. Populated via spatial join with the georef_id.

created_at	TIMESTAMPZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the farm record was created.
updated_at	TIMESTAMPZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the farm record was updated.
Table: tbl_insurance_record				
Field Name	Data Type	Length	Constraint	Description
insurance_record_id	SERIAL	50	PRIMARY KEY	Unique identifier for the insurance contract.
farmer_id	INT	50	FOREIGN KEY	References tbl_farmers_profile.farmer_id. Connects the farmer to the insurance record.
farm_id	INT	50	FOREIGN KEY	References farms.farm_id. Connects the policy to the land.
program_type	VARCHAR	100	NOT NULL	Type of insurance program.
product_name	VARCHAR	150	NOT NULL	Specific name of the insurance product.
effectivity_date	DATE	-	NOT NULL	Start date of the insurance coverage.
expiry_date	DATE	-	NOT NULL	End date of the insurance coverage.
amount_cover	DECIMAL	15, 2	NOT NULL	The maximum financial payout amount.
created_at	TIMESTAMPZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the insurance record was created.
updated_at	TIMESTAMPZ	-	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When the insurance record was updated.
Table: tbl_risk_assessments				
Field Name	Data Type	Length	Constraint	Description

assessment_id	SERIAL	-	AUTO_INCREMENT, PRIMARY KEY	Auto-incrementing unique ID for the report.
insurance_record_id	INT	50	FOREIGN KEY	References tbl_insurance_record.i nsurance_records_id. Links report to the contract.
summary_id	INT	4	FOREIGN KEY	References tbl_area_exposure_su mmmary.summary_id Links.
matrix_id	INT		FOREIGN KEY	References tbl_recsap_matrix.mat rix_id.
crop_stage_no	INT		NOT NULL	Numerical indicator of the crop's growth stage.
crop_stage	VARCHAR	150	NOT NULL	Text description of the growth stage
period_of_exposure	INT		NOT NULL	The calculated age of the crop in days at the time of the typhoon, derived by subtracting the effectivity_date from the typhoon occurrence date.
wind_velocity	INT		NOT NULL	The exact wind speed (in km/h) that the crop was subjected to.
indemnity_factor	DECIMAL	5, 4		The multiplier pulled from the look-up table based on crop type, stage, and wind velocity.
estimated_damage	DECIMAL	15, 2	NOT NULL	The calculated financial loss or damage estimate.

final_indemnity_p ayment	DECIMAL	15, 2	NOT NULL	The final calculated monetary payout required for the Finance Division report (amount_cover * indemnity_factor).
assessment_date	TIMESTAMPTZ	-	NOT NULL	Necessary for tracking when the calculation/assessment was performed, ensuring accurate record-keeping for management review and payout approval
user_id	INT		FOREIGN KEY	Necessary for tracking user ensuring accurate record-keeping for user.

Table: tbl_typhoons

Field Name	Data Type	Length	Constraint	Description
typhoon_id	SERIAL		AUTO_INCREMENT, PRIMARY KEY	Internal auto-incrementing unique ID utilized for system-level database joins and relational data management.
name	VARCHAR	100	NOT NULL	The assigned name of the storm (e.g., "Leon").
year	INT	4	NOT NULL	The year the storm occurred (e.g., 2026), preventing issues when PAGASA recycles names.
is_active	BOOLEAN		NOT NULL, DEFAULT TRUE	Master switch. When the storm leaves the PAR, this flips to FALSE and triggers the final payout

				computation.
Table: tbl_tropical_cyclone_bulletins				
Field Name	Data Type	Length	Constraint	Description
tcb_id	SERIAL	-	AUTO_INCREMENT, PRIMARY KEY	Auto-incrementing unique ID for the bulletin.
typhoon_id	INT		FOREIGN KEY	References tbl_typhoons.typhoon _id for the bulletin.
title	VARCHAR	255	NOT NULL	Full official title of the bulletin.
bulletin_count	INT	10	NOT NULL	the sequential number of the bulletin (e.g., 1, 2, 3).
category	VARCHAR	100	NOT NULL	Storm classification (e.g., "Tropical Depression").
max_sustained_wi nds	INT	10	NULLABLE	The actual maximum sustained wind speed near the center (typically stored as km/h).
gustiness	INT	10	NOT NULL	The actual maximum wind gusts (typically stored as km/h).
issued_at	TIMESTAMP TZ		NOT NULL	Timestamp when the bulletin was officially published.
expires_at	TIMESTAMP TZ		NOT NULL	Timestamp when the current bulletin is no longer valid.
center_geom	geometry	Point, 4326	NOT NULL	PostGIS spatial data representing the storm's eye/center.
Table: tbl_tcb_signals				
Field Name	Data Type	Length	Constraint	Description
signal_id	SERIAL	-	AUTO_INCREMENT, PRIMARY KEY	Auto-incrementing unique ID for the signal record.

tcb_id	INTEGER	4	FOREIGN KEY	References tbl_tropical_cyclone_b ulletins.tcb_id. Links warning of the storm.
signal_level	INTEGER	4	NOT NULL	The TCWS warning level (e.g., 1, 2, 3, 4, 5).
island_group	INTEGER	4	NOT NULL	Regional grouping code (0 = Luzon, 1 = Visayas, 2 = Mindanao).
area_name	VARCHAR	100	NOT NULL	The text name of the affected province/island.
Table: tbl_area_exposure_summary				
Field Name	Data Type	Length	Constraint	Description
summary_id	SERIAL		AUTO_INCREMENT, PRIMARY KEY	Unique identifier for this specific area-storm summary.
typhoon_id	INTEGER		FOREIGN KEY	References tbl_typhoons.typhoon _id. Identifies which storm caused this exposure.
boundary_id	INT		FOREIGN KEY	References tbl_admin_boundaries. boundary_id for actual area referencing.
province	VARCHAR	100	NOT NULL	The administrative province name (e.g., "Misamis Oriental").
municipality	VARCHAR	100	NOT NULL	The specific municipality or city name (e.g., "Talakag").
max_signal_level	INTEGER	1	NOT NULL	The highest (peak) TCWS level recorded for this area during the storm.

start_time	TIMESTAMPZ		NOT NULL	The issued_at timestamp of the first bulletin that placed this area under a signal.
end_time	TIMESTAMPZ		NOT NULL	The expires_at timestamp of the last bulletin that included this area.
is_eligible_6hr	BOOLEAN		DEFAULT FALSE	The Trigger: Automatically set to TRUE if max_signal_level >= 2 AND total_exposure_hours >= 6.0.
total_exposure_hours	DECIMAL	5, 2	NOT NULL	The mathematical sum of all active signal windows ($\sum$ expires - issued).
computed_at	TIMESTAMPZ		DEFAULT NOW()	Timestamp of when this summary was generated by the backend.
Table: tbl_admin_boundaries				
Field Name	Data Type	Length	Constraint	Description
boundary_id	SERIAL	-	PRIMARY KEY	Auto-incrementing unique ID for the map feature.
psgc_code	VARCHAR	10	UNIQUE, NOT NULL	The official 10-digit Philippine Standard Geographic Code. Prevents duplicate entries and spelling errors.

area_name	VARCHAR	100	NOT NULL, PARTIAL UNIQUE	Standardized name. Constrained by a Partial Unique Index (idx_unique_province_na me) to enforce uniqueness ONLY when boundary_type = 'Province'.
boundary_type	ENUM	-	NOT NULL	Classification level. Must be: 'Region', 'Province', or 'Municipality'.
parent_boundary_id	INT	-	FOREIGN KEY	Self-referencing link to boundary_id. Links a Municipality to its Province. Left NULL if the record is a Region or Province.
geom	GEOMETRY	MultiPoly gon, 4326	NOT NULL	PostGIS spatial data representing the area's official administrative boundary.

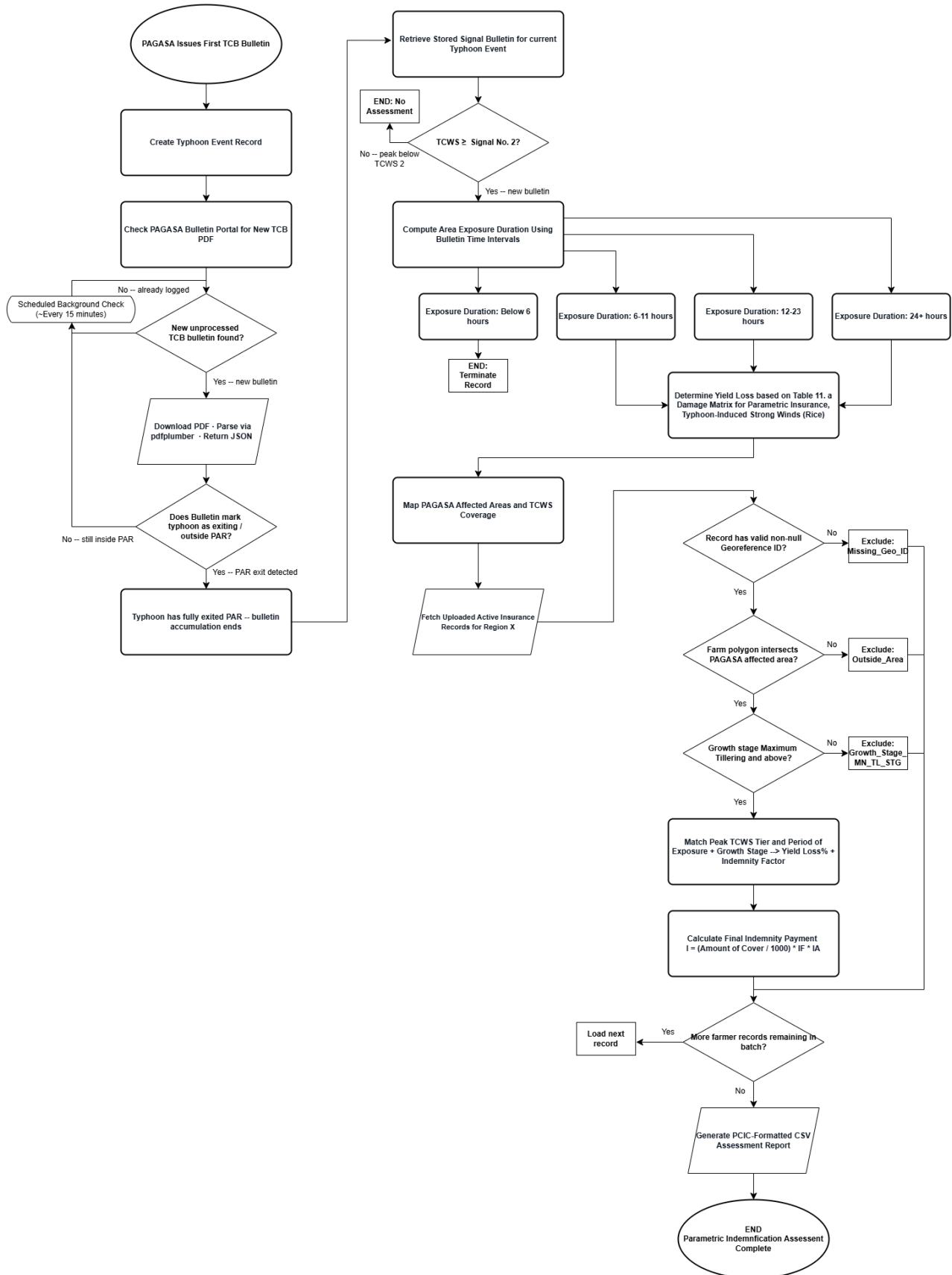
Table: tbl_recsap_matrix

Field Name	Data Type	Length	Constraint	Description
matrix_id	SERIAL	-	AUTO_INCREMENT, PRIMARY KEY	Unique identifier for this specific parametric rule.
crop_stage_no	INT	-	NOT NULL	Numerical indicator of the crop's growth stage (0 through 9).
wind_signal_tcw s	INT	-	NOT NULL	The PAGASA Tropical Cyclone Wind Signal level (1 through 5).
estimated_yield_ loss	DECIMAL	5, 2	NOT NULL	Standard expected loss percentage (e.g., 20.00, 30.00) for UI display.
indemnity_factor	DECIMAL	5, 4	NOT NULL	The strict multiplier (e.g., 0.2640, 0.3920,

				1.0000) used by the backend to compute the final payout.
is_active	BOOLEAN	-	NOT NULL, DEFAULT TRUE	Allows PCIC to update rates in the future without breaking historical claims.

Table 2. Database Dictionary

3.3.6 Parametric Indemnity Process Flowchart



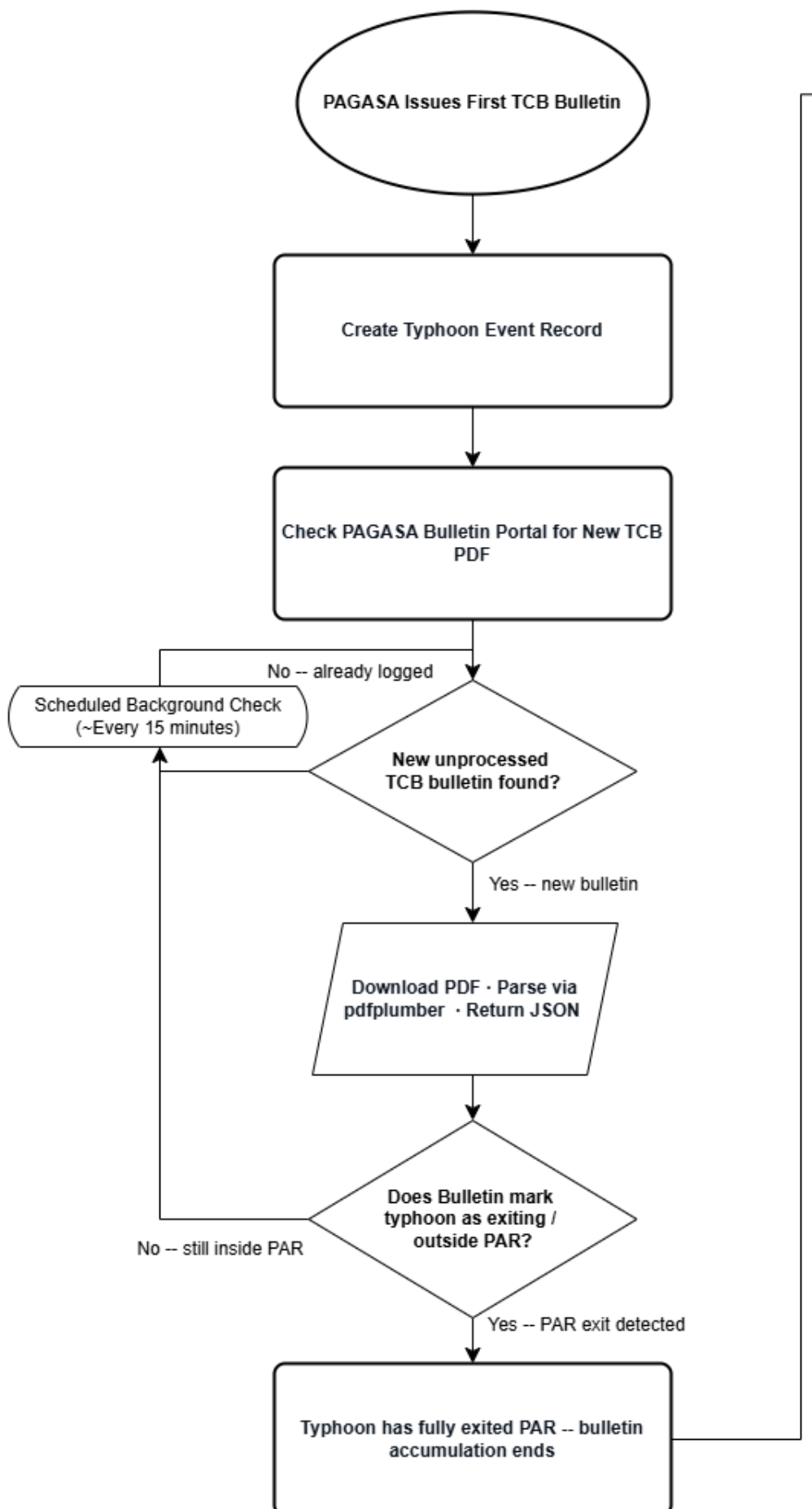
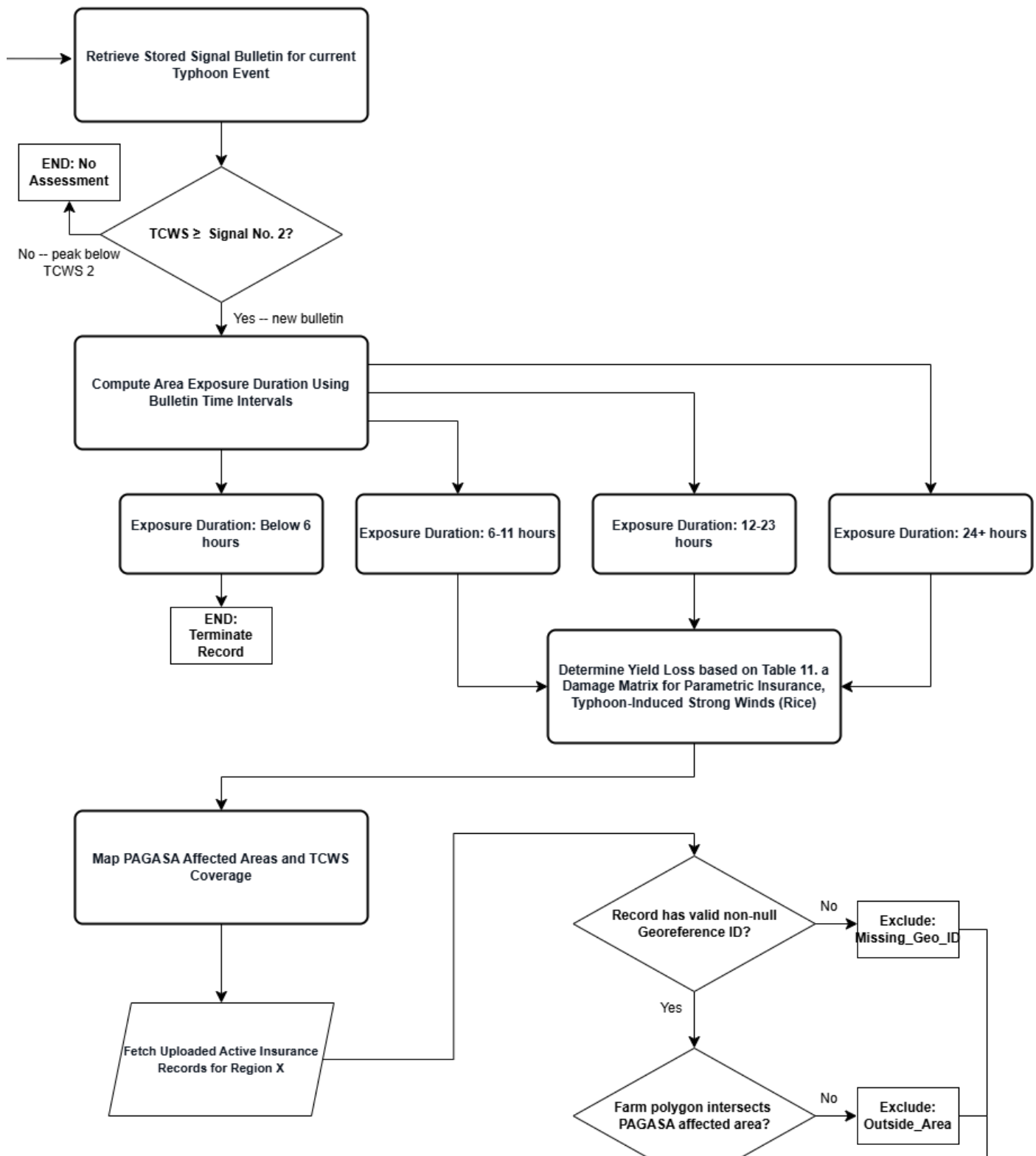
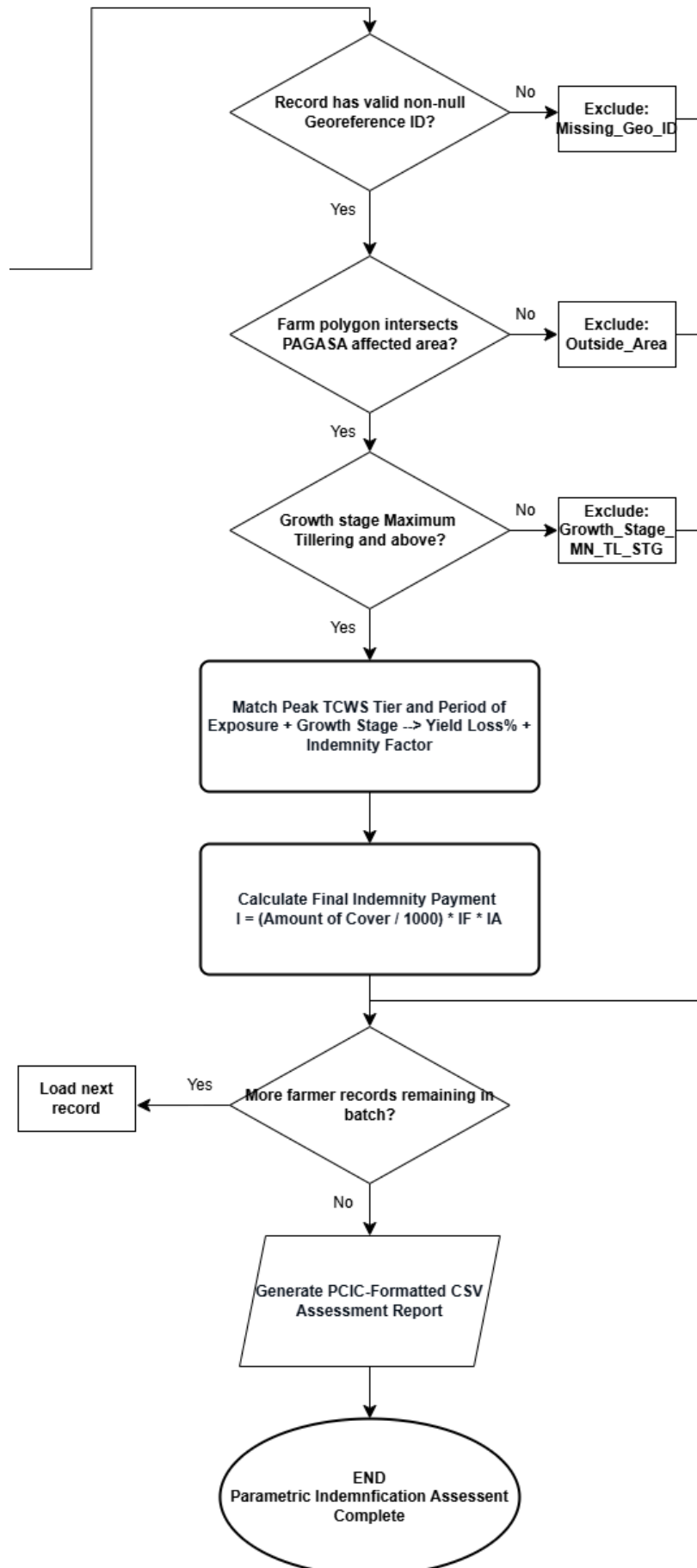


Figure 8. Parametric Indemnity Process Flowchart





Traditional agricultural insurance mainly depends on individual field assessments, where insurance adjusters personally inspect damaged farms to determine crop losses. This traditional method often leads to high administrative expenses, inconsistent evaluations, and long processing periods that delay financial assistance for affected farmers.

To improve the claims process, the Philippine Crop Insurance Corporation introduced parametric or index-based insurance. Under the Revised Claims Settlement Approaches and Procedures (ReCSAP), the claims process for typhoon-affected farms becomes more objective and technology-based. When a tropical cyclone enters the region, the PCIC Automated Business System uses Geographic Information System technology to compare real-time weather data with the exact georeferenced locations of insured rice farms.

For strong wind hazards, damage assessment is based entirely on a predefined Damage Matrix. Instead of conducting physical field inspections, the system processes three important parameters:

1. **Meteorological Severity** – the official Tropical Cyclone Wind Signal (TCWS) issued by PAGASA.
2. **Period of Exposure** – the total number of hours the farm is exposed to strong winds.
3. **Biological Vulnerability** – the growth stage of the rice crop during the typhoon event.

The computation of a parametric insurance claim under the ReCSAP framework uses the Comprehensive Parametric Indemnity Formula.

Since the Amount of Cover (AC) is assigned on a per-hectare basis, the total farm area must also be included in the computation of the final claim amount.

$$I = (AC / 1000) \times IF \times Area$$

Where:

- I = Total indemnity payout in Philippine Pesos
- AC = Amount of Cover per hectare
- IF = Indemnity Factor based on yield loss and crop growth stage
- Area = Total insured land area in hectares

The step-by-step computation process involves:

Step 1: Verification of Crop Growth Stage

Based on the official parametric damage matrix, the insured crop must belong to one of the eligible growth stages:

- Booting
- Flowering
- Maturity

For example, a rice crop is verified under the **Flowering Stage**, which is considered highly vulnerable to strong winds because it may cause spikelet sterility and flower damage.

Step 2: Determine the Yield Loss Percentage

The following environmental and crop conditions are used in the computation:

- **Wind Signal:** TCWS No. 3 (89–117 kph)
- **Exposure Duration:** 24 hours
- **Growth Stage:** Flowering Stage

Period of Exposure (Hours)	Growth Stage		
	Tropical Cyclone Wind Signal No.02 (62-88 KPH)		
	BOOTING	FLOWERING	MATURITY
	Yield Loss	Yield Loss	Yield Loss
6	10	15	<10
12	15	20	10
24	20	25	15
Period of Exposure (Hours)	Tropical Cyclone Wind Signal No.03 (89-117 KPH)		
6	15	20	10
12	20	25	15
24	25	30	20

Period of Exposure (Hours)	Tropical Cyclone Wind Signal No.04 (118-184 KPH) AND > 184 KPH		
6	20	25	15
12	25	30	20
24	30	35	25

Based on the official damage matrix, the intersection of these values results in a yield loss percentage of

$$YL\% = 30\%$$

Step 3: Identification of the Indemnity Factor

Using the Rice Indemnity Factor Table:

- The crop belongs to the Reproductive Stage because it is in the flowering phase.

The corresponding Indemnity Factor is:

- The yield loss category falls under greater than 25% up to 30%

YIELD LOSS %	EARLY VEGETATIVE	LATE VEGETATIVE	REPRODUCTIVE	LATE REPRODUCTIVE	MATURITY
>10 to 15	146	170	194	218	243
>15 to 20	198	231	264	297	330
>20 to 25	248	289	330	372	413
>25 to 30	294	343	392	441	490
>30 to 35	336	392	448	504	560

$$IF = 392.00$$

Step 4: Final Indemnity Payout Calculation

The operational values used are:

- Amount of Cover: ₱25,000.00 per hectare
- Indemnity Factor: 392.00
- Total Land Area: 2.87 hectares

Substituting these values into the formula:

$$I = (25,000 / 1000) \times 392.00 \times 2.87$$

First, simplify the amount of cover ratio:

$$I = 25 \times 392.00 \times 2.87$$

Compute the indemnity value per hectare:

$$I \text{ per hectare} = 25 \times 392.00 = 9,800.00$$

Finally, multiply the result by the total farm area:

$$I = 9,800.00 \times 2.87$$

Final computation:

$$I = 28,126.00$$

Therefore, the final indemnity payout is **₱28,126.00**.

3.4. System Development

3.4.1 Hardware Requirements

Hardware Component	Desktop Minimum	Desktop Recommended	Laptop Minimum	Laptop Recommended
Processor (CPU)	Intel Core i5 (11th Gen) or AMD Ryzen 5	Intel Core i7 (12th Gen) or AMD Ryzen 7	Intel Core i5 (11th Gen) or AMD Ryzen 5	Intel Core i7 (12th Gen) or AMD Ryzen 7
Memory (RAM)	16 GB DDR4	32 GB DDR4/DDR5	16 GB DDR4	32 GB DDR4/DDR5
Storage	256 GB SSD	512 GB to 1 TB SSD	256 GB SSD	512 GB to 1 TB SSD
Graphics (GPU)	Integrated Intel Iris Xe / AMD Radeon Graphics	Dedicated NVIDIA RTX 3060 / 4060 Mobile	Integrated Intel Iris Xe / AMD Radeon Graphics	Dedicated NVIDIA RTX 3060 / 4060 Mobile

Table 3. Hardware Requirements

To effectively and efficiently develop and deploy AgriSureGIS, the hardware framework was established based on the desktop infrastructure used by PCIC GIS specialists and the laptops used by the proponents. This setup provides the minimum baseline and

recommended upgrades required to run the web-based GIS application, host the localized database, integrate Google Earth Engine, and render massive farm polygon datasets without performance bottlenecks.

3.4.2 Software Requirements

The AgrisureGIS system will utilize a modern and scalable software stack designed to support web-based geospatial map visualization, automated parametric crop damage assessment, remote sensing-based farm verification, automated meteorological data scraping, and administrative user management. The system will include dynamic web-based dashboard functionalities to provide GIS specialists and system administrators with a unified and efficient agricultural risk assessment and insurance verification platform.

To support the modular, three-tier architecture of AgrisureGIS, the following software tools and frameworks will be utilized:

Web Application Development

The core development of the AgrisureGIS web interface is engineered utilizing React and Tailwind CSS to deliver a highly modular and responsive user experience. React serves as the foundational component-based frontend library, managing client-side state transitions dynamically to allow real-time updates and seamless interactivity without full-page reloads. Tailwind CSS is fully integrated into this presentation layer, leveraging its utility-first design classes to construct a fluid, cross-browser responsive layout optimized for data-heavy administrative layouts.

Web-Based Spatial Dashboard

The interactive, map-centric capabilities of the platform are driven by the integration of Leaflet JS and GeoServer. Leaflet JS functions as the primary client-side mapping engine, embedded within the frontend architecture to render interactive map viewports, handle base imagery layers, and dynamically map clickable farm vector data. Bridging the gap between storage and display, GeoServer acts as the dedicated map server, establishing direct connections to the spatial database to translate geographic attributes and stream them onto the Leaflet canvas using standardized, high-performance Web Map Service (WMS) and Web Feature Service (WFS) protocols.

File Ingestion and Extract, Transform, Load (ETL) Pipeline

To automate the processing of real-time hazard data, the system implements an automated ETL pipeline within the FastAPI application layer. The system initiates data acquisition using the Requests and BeautifulSoup4 libraries to programmatically crawl the PAGASA portal and retrieve the latest active Tropical Cyclone Bulletin PDF. Data extraction from the document structure is handled via pdfplumber, parsing tabular wind speeds, meteorological coordinates, and structural text areas. The extracted data is structurally validated using Pydantic data models and serialized into a structured JSON format. Finally, database ingestion into the localized PostgreSQL/PostGIS database is managed asynchronously via SQLAlchemy coupled with GeoAlchemy2 for handling spatial constraints, ensuring object-relational mapping integrity and preventing operational data loss.

To accommodate the importation of compressed or loose multi-sourced files from the PCIC, the application layer utilizes an automated file processing pipeline written in Python and executed within FastAPI. File extraction, workspace mapping, and system background decompression are driven natively by the integrated utilization of the Python zipfile and pathlib standard libraries, while boundary duplicate checking is managed through cryptographic validation filters powered by hashlib to verify payload uniqueness before deep record processing. Alphanumeric record transformation is achieved by integrating the pandas data manipulation framework to parse, clean, and isolate structural attributes from unzipped .csv registries. In parallel, geometric structures are constructed by parsing raw Extensible Markup Language (XML) data streams from field-captured .gpx tracks using the gpxpy extraction library.

To ensure topological validity, these separate spatial indicators are mapped within the GeoPandas framework, where low-level geometric engines driven by Shapely enforce strict validation rules to weld individual waypoints into valid, uncorrupted geographic boundaries (POLYGON or MULTIPOLYGON features). To align diverse coordinates with web mapping layers and weather footprints, a coordinate transformation component driven by pyproj performs mathematical projection mapping, standardizing all spatial layers to the global WGS 84 (SRID 4326) reference datum. Data persistence and Object-Relational Mapping (ORM) transactions are maintained by converting the parsed GPX streams directly into native geometry types, leveraging SQLAlchemy to stream alphanumeric metadata records alongside

GeoAlchemy2 to map and insert these spatial features directly into persistent PostgreSQL/PostGIS geometry columns.

Backend and Spatial API Services

The core processing logic and application routing of the AgriSureGIS platform are engineered using FastAPI, which serves as a high-performance, asynchronous backend framework capable of managing concurrent HTTP requests without bottlenecking complex spatial operations. This application layer orchestrates communication between the client-side dashboard and the database via RESTful API endpoints, allowing for seamless data retrieval, dynamic spatial filtering, and the execution of the automated parametric damage assessments.

To manage the intricate relationship between geographic boundaries and alphanumeric farmer profiles, which are structurally aligned with the Philippine Standard Geographic Code (PSGC) mandated by the Philippine Statistics Authority (PSA), the backend interfaces directly with a localized PostgreSQL database enhanced by the PostGIS extension. PostGIS provides the specialized spatial indices and geometric functions required to store, query, and enforce strict mapping constraints on complex coordinate data using the global WGS 84 (SRID 4326) reference system. Furthermore, bridging the application layer and the spatial database, SQLAlchemy and GeoAlchemy2 are implemented for Object-Relational Mapping (ORM). SQLAlchemy abstracts raw SQL transactions and securely manages database connections, while GeoAlchemy2 extends this ORM capability to natively support PostGIS spatial data types, ensuring the seamless translation of multi-polygon features between the application code and database geometry columns.

Additionally, to facilitate satellite-based crop verification beneath intense storm cloud cover, the backend architecture incorporates the Google Earth Engine Python API as an external cloud-side geospatial service. The FastAPI application programmatically authenticates with GEE using secure service account credentials to query the planetary-scale, cloud-hosted Sentinel-1 Synthetic Aperture Radar Ground Range Detected (COPERNICUS/S1_GRD) dataset based on the localized PostGIS farm bounding coordinates and target typhoon timeline parameters. By leveraging server-side geoprocessing, the backend completely offloads multi-gigabyte radar computation, speckle filtering, and radiometric calibration to Google's remote cloud infrastructure. The API then returns these pre-processed

radar backscatter data streams directly to the application layer as lightweight map tiles or structured spatial vectors, enabling the instant, web-based visual validation of crop presence by the authorized GIS specialist without exhausting local hardware memory resources.

Automated Reporting and Institutional Interoperability

To fulfill the operational requirement of accelerating the indemnification process, the architecture incorporates a dedicated reporting and data export module. Once the parametric damage assessment is finalized, the spatial intersections and computed financial multipliers are aggregated within the FastAPI backend. The system utilizes the pandas data manipulation framework to dynamically structure these results into a standardized tabular data frame. This programmatic compilation ensures that the exported records strictly conform to the legacy CSV schema required by the PCIC Finance Division for payout approval. The backend then securely transmits this structured data payload to the React frontend, which initiates a localized browser download. This seamless data transition guarantees that the final assessment reports are highly interoperable, immutable, and immediately accessible for institutional review without requiring secondary manual transcription.

Baseline Data Acquisition and System Initialization

To establish the foundational geographic and administrative parameters of the system, territorial boundaries and official region codes are treated as static baseline data. Because Philippine municipal borders and the Philippine Standard Geographic Code (PSGC) update infrequently, the architecture intentionally avoids the overhead of automated API pipelines for these datasets. Instead, during the initial system deployment on the localized PCIC server, the System Administrator performs a one-time manual database seeding process. Official alphanumeric PSGC masterlists acquired from the Philippine Statistics Authority (PSA), alongside mathematically precise administrative shapefiles sourced from the Humanitarian Data Exchange (HDX), are imported directly into the on-premise PostgreSQL/PostGIS database. This manual initialization strategy eliminates unnecessary external web dependencies, minimizes application complexity, and leverages the secure, direct-access capabilities of the localized server infrastructure, ensuring the system is fully operational and territorially accurate prior to handling dynamic typhoon events.

Real-Time Notification and Alert System

To ensure synchronous communication and align with the system's data flow architecture, AgriSureGIS implements a real-time notification mechanism to alert the GIS Specialist of critical system events, such as automated PAGASA extraction failures and the successful parsing of new bulletins. To complement in-app alerts, an automated email dispatch subsystem configured via SMTP within the FastAPI backend is triggered to instantly notify the GIS Specialist's registered inbox whenever a new Tropical Cyclone Bulletin (TCB) is successfully extracted and processed. The core dashboard process is driven by WebSockets integrated within the FastAPI backend, which establishes a persistent, bi-directional communication channel with the React frontend. When an exception or operational milestone is triggered within the backend pipeline, a serialized event payload is pushed directly to the client layer, concurrently executing the email service for remote awareness. The frontend captures this data stream and utilizes React to render non-intrusive, context-aware alert modules on the dashboard interface. This automated, dual-channel feedback loop guarantees that administrative users receive immediate, localized system warnings and external email updates without requiring manual page refreshes, thereby maintaining operational continuity and a seamless flow of data between the application tiers.

Failure Handling and Rollback Mechanisms

In addition to protecting the database from crashes, the system is designed to handle human errors and outside service limits smoothly. First, if a GIS Specialist uploads a broken spreadsheet, such as a CSV file with missing columns or incorrect text formats, the system will not reject the entire file. Instead, it checks the data first, saves all the correct rows, and points out exactly which lines the user needs to fix. Second, when the system checks thousands of farms at once, Google Earth Engine might temporarily block it for sending too many requests too quickly. To prevent the process from failing, the system uses an automatic "pause and retry" mechanism that slowly spaces out the requests until Google processes all of them. Third, if the automated system fails to download or read the PAGASA Tropical Cyclone Bulletin, whether due to a website outage or an unexpected change in the PDF format, the application will not crash. Instead, it immediately alerts the GIS Specialist and allows them to manually upload the bulletin file so the damage assessment can continue without delay. Finally, if the local map server is struggling to load heavy farm boundaries, the web dashboard is programmed so it will not freeze. It will keep trying to load the maps in the

background, allowing the user to continue clicking around and using other parts of the application without interruption.

Development and Environment Tools

- Visual Studio Code (VS Code): serves as the primary Integrated Development Environment (IDE) and code editor, which is utilized for managing and writing the scripts across your entire code repository.
- Python Runtime Environment: serves as the foundational execution environment for the backend architecture. It interprets and executes the asynchronous server logic, automation scripts, and geometric processing pipelines.
- Node.js + Vite: functions as the frontend build tool and package manager, designed for efficiently bundling and compiling JavaScript or React mapping libraries.
- Git & GitHub act as the version control system, providing the necessary infrastructure for managing code changes, utilizing branching features, and facilitating team collaboration.
- React Leaflet / OpenLayers: The JavaScript-based spatial visualization libraries integrated into the React frontend. These tools render the interactive map view on the client dashboard, allowing the GIS Specialist to see the overlaid PAGASA storm paths and farmer farm polygons directly in the web browser.
- pgAdmin 4: ensure that the SQLAlchemy and GeoAlchemy2 database transactions succeeded while simultaneously confirming that the geometry layer displays accurately on the tool's built-in map viewer.
- pytest + HTTPX: serves as the system's asynchronous integration and functional testing framework, used to programmatically verify the correctness of the FastAPI backend.
- FastAPI: Functions as the high-performance, asynchronous web framework used to build the system's API endpoints, handling request routing and Pydantic data validation.
- GeoServer: Functions as the open-source geospatial data server. It acts as the map publishing environment that connects to PostGIS, processes geographical vectors, and serves map layers as OGC-compliant Web Map Services or Web Feature Services.

3.5. System Testing and Evaluation

3.5.1 Functionality Testing

Functionality testing will be conducted by the researchers and authorized PCIC GIS specialist personnel to verify the system's ability to accurately ingest, process, and clean incoming data packages under a unified backend workflow. Testing will verify the platform's capability to intercept loose multi-file submissions of tabular records and geographic coordinates, alongside bulk compressed archives, ensuring both entry points normalize data identically. Perimeter security checks will verify the application's ability to calculate cryptographic content hashes, confirming that duplicate user files are caught and rejected before triggering database transactions.

Additional validation testing will be conducted to verify the system's ability to execute cross-runtime tasks when capturing real-time meteorological indicators. By injecting mock weather bulletins with known text parameters, testing will confirm that pattern-matching scripts parse cyclone metadata, local wind risk signals, and forecast coordinates into structured data matrices and are extracted with exact precision.

Geospatial processing audits will be conducted to verify the system's ability to convert raw global positioning coordinates directly into spatial database geometries standardized to a uniform global datum. This phase will evaluate execution rules to confirm that the extracted Tropical Cyclone Wind Signal hazard zones derived directly from the parsed bulletin data successfully intersect target farm boundaries without data loss. Finally, structural resilience testing will be conducted to verify the system's ability to handle runtime anomalies, ensuring that unexpected script errors or API network drops instantly trigger database rollbacks and storage purges to protect baseline server memory.

3.5.2 Parametric Calculation and Spatial Accuracy Testing

Evaluating the computational precision of the disaster assessment engine requires strict validation of the algorithmic logic and spatial join operations that determine insurance payouts. Algorithmic compliance testing verifies that the system correctly translates complex relational logic rules natively at the database level by cross-referencing localized crop growth stages against meteorological wind signals to isolate exact payout percentages from reference matrices.

To ensure mathematical precision during large-scale automated assessments, spatial intersection audits test the processing layer against known control datasets. This validation method uses fixed geometric overlaps and simulated buffer shapes to check that the calculated loss area matches exact geographic realities, ensuring that overlapping spatial features or topological edge cases do not trigger calculation errors or misaligned risk ratings. Furthermore, data flow tracking exercises monitor the movement of values from the initial hazard calculation through to final database logging. This verifies that individual farmer records, audited loss percentages, and finalized financial multipliers map cleanly across relational database tables without discrepancies, guaranteeing mathematical accuracy across all processing cycles.

3.5.3 System Usability Scale Testing

System Usability Scale Questionnaire	Strongly Disagree				Strongly Agree
1. I think that I would like to use this product frequently.	1	2	3	4	5
2. I found the product unnecessarily complex.	1	2	3	4	5
3. I thought the product was easy to use.	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this product.	1	2	3	4	5
5. I found the various functions in the product were well integrated.	1	2	3	4	5
6. I thought there was too much inconsistency in this product.	1	2	3	4	5
7. I imagine that most people would learn to use this product very quickly.	1	2	3	4	5
8. I found the product very awkward to use.	1	2	3	4	5
9. I felt very confident using the product.	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this product.	1	2	3	4	5

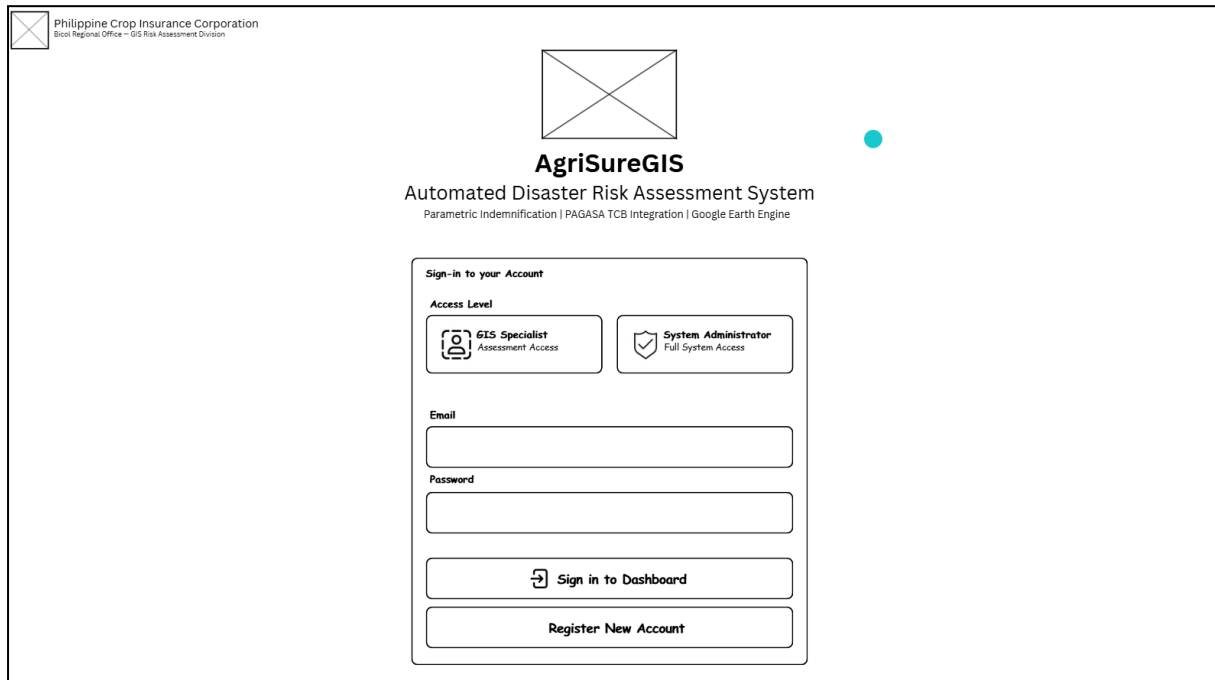
Table 4. SUS Questionnaires

A standardized System Usability Scale (SUS) is utilized as the primary quantitative instrument. The system will be rigorously tested and evaluated by the authorized PCIC GIS Specialist and the Division Chief of the PCIC Region X. These specific evaluators were selected to represent both the hands-on technical workflow and the high-level administrative oversight. The instrument consists of 10 alternating positive and negative statements scored on a 5-point Likert scale, ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). This structure minimizes response bias and yields a reliable global usability score (0% to 100%)

to mathematically verify if the system effectively reduces the time and friction involved in post-typhoon crop damage assessments.

3.6. Wireframe

3.6.1 Low Fidelity Wireframe



Philippine Crop Insurance Corporation
Bicol Regional Office – GIS Risk Assessment Division



AgriSureGIS
Automated Disaster Risk Assessment System
Parametric Indemnification | PAGASA TCB Integration | Google Earth Engine

Sign-in to your Account

Access Level

 **GIS Specialist**
Assessment Access

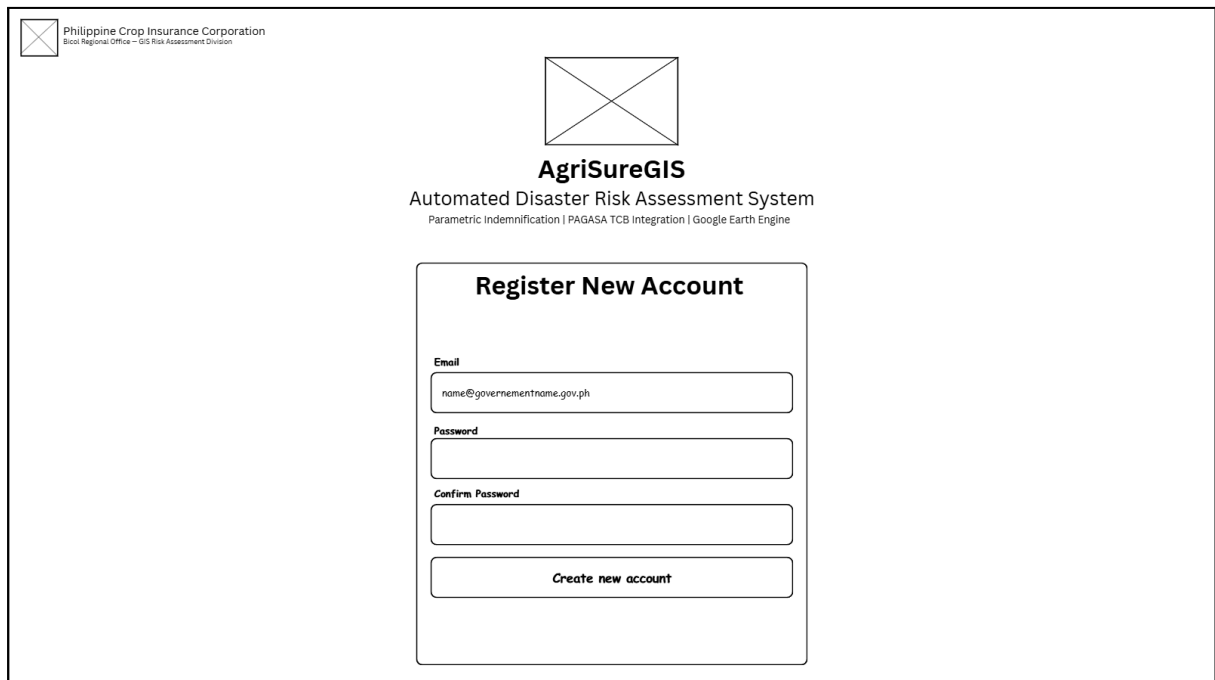
 **System Administrator**
Full System Access

Email

Password

 **Sign in to Dashboard**

Register New Account



Philippine Crop Insurance Corporation
Bicol Regional Office – GIS Risk Assessment Division



AgriSureGIS
Automated Disaster Risk Assessment System
Parametric Indemnification | PAGASA TCB Integration | Google Earth Engine

Register New Account

Email

Password

Confirm Password

Create new account

Login & Register Page

PCIC-GIS
Risk Assessment System

Monitoring Spatial Assessment Settings

Parser Active

Filter Map: All

Export Period of Exposure: SAR/REE Analysis

SAR Analysis - Area of Interest
Export from Time Interval: 1 Day

Area of Interest

Municipality: By Municipality

REE Configuration

Date Range

Analysis Type

Submit SAR Request to REE

Growth Stage
Seeding
Vegetative
Reproductive
Ripening
Not Planted

Signal No.
Signal 1
Signal 2
Signal 3

Former Records

Former ID	Insured Name	Municipality	Barangay	Farm ID	Area(ha)	Planted	Plant Date	Stage	Signal	Exp(h)

Drop files here or click to browse

Monitoring Page with Notification Feature

PCIC-GIS
Risk Assessment System

[Monitoring](#)
[Spatial](#)
[Assessment](#)
[Settings](#)

Parser Active

 Username

Active Typhoon

Typhoon Name

TCB Downloaded

6

Affected Farms

19/20

EST. Total Indemnity

₱ 380K

PAGASA TCB BULLETIN Typhoon Name

Bulletin	Issued	Signal	Wind Speed	Affected Areas	Status	Action
TCB No. 1	01 Nov 2024 00:00 PHT	No. 1	30-60 kph	Camarines Norte, Camarines Sur..	Processed	

System Status

PAGASA Parser
GEE Connection
Email Alerts
Database Backup
Session Monitor

Last check: 15:05 PHT

Growth Stage Distribution

Seedling (5) Vegetative (6) Reproductive (5) Ripening (4)

PCIC-GIS
Risk Assessment System

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[Spatial](#)
[Assessment](#)
[Settings](#)

Parser Active

 Username

Active Typhoon

Typhoon Name

TCB Downloaded

Affected Farms

EST. Total Indemnity

PAGASA TCB BULLETIN Typhoon N

Bulletin	Issued
TCB No. 1	01 Nov 2024 00:00 P

TCB No. 1 — Tropical Cyclone Typhoon Pepito

PAGASA-DOST - 01 Nov 2024 00:00 PHT

**Philippine Atmospheric, Geophysical
And Astronomical Services Administration**

PAGASA — Tropical Cyclone Bulletin

Public Storm Warning Signal

No. 1

30-60 kph

Bulletin No.

Issued By:

Wind Velocity:

Version:

Cyclone Name:

Date / Time:

File Size:

Status:

Areas Under Storm Warning Signal No.

- Camarines Norte
- Camarines Sur
- Catanduanes
- Albay

Close

Download TCB

Distribution

Active (6) Reproductive (5) Ripening (4)

Monitoring and Extraction Module



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Risk Assessment System

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[Spatial](#)
[Assessment](#)
[Settings](#)

Parser Active




 Username

Typhoon Events

Select a TCB to trigger Assessment

TCB No. 1
01 Nov 202
30-60 kph
Processed

Assessment Reporting | Typhoon name

Filtered Records
10

Signal 2 Planted
10

Total Area
24.0ha

Total Sum Insured
P600k

Total Indemnity
P188K

Farmer ID	Name	Municipality	Farm ID	Area	Stage	Signal	Exp(h)	Wind(kph)	Sum(ins)	Ind. %	Payment
00001	Juan A. Dela Cruz	Naga City	F001	2.50	Vegetative	2	6	61-120	P62,500	I25%	P15,625
Total:				00.00					P123,456		P123,456

Indemnity by Municipality
Signal 2 farms only

Naga
Pili
Libanan
Sipocot
Gua
Lagonoy

Damage Factors
Signal 2 farms only

Stage
Seedl
Veget
Repro
Ripen

Signal 2 Farms (Planted)
10 farms
P187,500 est. indemnity



PCIC-GIS
Risk Assessment System

[Monitoring](#)
[Spatial](#)
[Assessment](#)
[Settings](#)

Parser Active




 Username

Typhoon Events

Select a TCB to trigger Assessment

TCB No. 1
01 Nov 202
30-60 kph
Processed

Step 1 – Review CSV Before Generating

PCIC Indemnification Report - Typhoon Pepito - TCB No. 1 - Camarines Sur

Please review all 10 records below before confirming download. This report covers only planted Signal No. 2 farms.

Row ID	Farmer ID	Name	Municipality	Barangay	Farm ID	Area	Planted	Planted Date	Growth Stage	Crop Type	Sum(ins)	Signal	Exp(h)	Wind(kph)	Ind. %	Payment
00001	PCIC-2024-00001	Juan A. Dela Cruz	Naga City	Triangulo	F001	2.50	Yes	2024-09-10	Vegetative	Rice	P62,500	2	6	61-120	25%	P15,625

1 records | Total SI: P123,456 | Total Indemnity: P123,456

Cancel
Step 2 - Confirm & Download CSV

Signal 2 Farms (Planted)
10 farms
P187,500 est. indemnity

Assessment and Reporting Module


PCIC-GIS
Risk Assessment System

 Monitoring
  Spatial
  Assessment
  Settings

• Parser Active
 


 Username


Calibration Settings
Save All Settings

Configure system parameters, credentials, and user access.

RSBA Coverage Rate (Sum Insured)
 RSBSA Rate — Rice (P per hectare)
 Default : 00000
 PCIC RSBSA fixed rate. Applies to all rice farmer records.

Live Preview — Impact at Current Rate

Rate per Hectare	Sample: 2.5 ha	At 40% ind. factor
P25,000	P62,500	P25,000

Per-record overrides can be applied in the Assessment module by clicking the edit icon on any row's Sum Insured column.


Signal Damage Factor Table

Edit the percentage damage factor applied per growth stage and signal number. Changes take effect on the next analysis run.

Growth Stage	Signal No. 1 (30-60 kph)	Signal No. 2 (61-120 kph)	Signal No. 3 (121-185 kph)
Seedling	%	%	%
Vegetative	%	%	%
Reproductive	%	%	%
Ripening	%	%	%


PCIC-GIS
Risk Assessment System

 Monitoring
  Spatial
  Assessment
  Settings

• Parser Active
 


 Username


Google Engine Credentials
Save All Settings

Configure Google Cloud Project credentials for SAR (Synthetic Aperture Radar) imagery analysis via Google Earth Engine.

GCP Project ID

EEC Capacity Allocated

Service Account API Key

Used to authenticate SAR imagery requests for crop area analysis within farm polygons.


PAGASA Parser & Notification Settings

TCB Polling Interval (hours)

Alert Email Recipients

How often to check the PAGASA website for new bulletins during an active typhoon.

Comma-separated email addresses notified when a new TCB is parsed.

Email Alerts
 Send email when new TCB is downloaded ☐

On-Screen Pop-up Alerts
 Show in-app notification on new bulletin ☐


PCIC-GIS
Risk Assessment System

[Monitoring](#)
[Spatial](#)
[Assessment](#)
[Settings](#)

Parser Active

Username


Session and Security Settings

Save All Settings

Session Timeout (minutes)

Auto-Save Work
Prevent data loss on session timeout
☐


Database Backup and Restore

Last Backup
30 Oct 2024

Backup Size
124 MB

Stored Backups
14 versions

Run Backup Now

Restore from Backup

☐ Automatic daily backup at 23:00 PHT


User Account Management

Manage system access for GIS specialists and administrators.

Add User

Name	Role	Email	Status	Actions
Ana L. Reyes	GIS Specialist	a.reyes@pcic.gov.ph	Active	Edit Delete

Ana L. Reyes
GIS Specialist
a.reyes@pcic.gov.ph
Account Settings
Sign Out

Calibration and Settings Module

3.7. Deployment

the system utilizes an on-premise deployment strategy rather than a public cloud infrastructure. The core backend services including the FastAPI server, GeoServer, and the PostgreSQL/PostGIS database are hosted centrally on a dedicated physical server located securely within the PCIC Region X office. The client-side WebGIS dashboard operates as a desktop application that connects to this central server strictly via the office's Local Area Network (LAN/Intranet). This localized architecture guarantees that multiple GIS specialists can simultaneously access, update, and assess synchronized spatial data without ever exposing sensitive farmer profiles to the public internet, effectively eliminating data silos while maintaining absolute institutional data sovereignty.

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